PLUS CONSTRUCTIONS AND CANONICAL INDUCTION FOR GREEN CATEGORIES

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Abstract. We develop canonical induction for Green categories associated with Mackey systems. We establish a change-of-rings framework in which the lower- and upper-plus constructions are realized by induction and coinduction of modules. The two plus adjunctions yield the mark morphism, and, when the relevant group orders are invertible in the coefficient ring, Möbius inversion gives an explicit inverse. Using the resulting linearization and tom Dieck maps, we construct a canonical induction map, give an explicit formula for it, and characterize when it is a right inverse of linearization. In the classical setting of Mackey functors, our construction and splitting criterion recover Boltje's results.

Contents

1. Introduction 1
2. Preliminaries 4
3. Green categories from Mackey systems 7
4. A quadruple of group categories 9
5. The Burnside idempotent decomposition 15
6. The plus constructions 17
7. The mark map and its inverse 24
8. Canonical induction 27
References 30

1. Introduction

Brauer's induction theorem states that every complex character of a finite group can be expressed as an integral linear combination of characters induced from linear characters of elementary subgroups. Boltje [Bol98] developed canonical induction in the general setting of Mackey functors. In this framework, the Mackey functor M plays the role of the character ring, while the restriction functor A plays the role of the linear characters. Unlike M, the functor A is equipped with restriction maps but not induction maps. The lower-plus construction produces a Mackey functor A_+ from A, together with an induction

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morphism $A_+ \to M$. Canonical induction seeks a restriction-compatible section to this morphism, and Boltje established criteria for when such a splitting exists. The upper-plus construction provides a complementary construction A^+, related to A_+ by the mark morphism $A_+ \to A^+$. This relationship plays a central role in constructing canonical induction formulas.

Coşkun [Coş07] later gave a module-theoretic interpretation of the plus constructions in terms of induction and coinduction. Boltje, Raggi-Cárdenas and Valero-Elizondo [BRCVE19] extended the lower- and upper-plus constructions to biset functors on suitable subcategories of the biset category. They also established the corresponding adjunctions and studied the mark morphism and the multiplicative structures of these constructions for Green biset functors. Further extensions include the fibered biset-functor constructions of Calderón [Cal25] and the categorification using Mackey 2-functors developed by Balmer and the author [BM26].

The purpose of the present paper is to study the plus constructions and canonical induction in the setting of Green categories associated with Mackey systems. Let $\mathfrak{K}$ be a set of finite groups closed under taking subgroups. A Mackey system on $\mathfrak{K}$ is a subcategory of the category of finite groups with object set $\mathfrak{K}$, whose morphisms satisfy the Mackey-system axioms. Following Barker [Bar16b], a pair $(\mathcal{I}, \mathcal{R})$ of Mackey systems satisfying the Green-category axioms defines a Green category, namely the $\mathbb{Z}$-linear subcategory of the biset category generated by induction along morphisms of $\mathcal{I}$ and restriction along morphisms of $\mathcal{R}$.

We concentrate on ordinary induction, namely induction along injective homomorphisms, while allowing restriction along the morphisms of an arbitrary Mackey system.

1.1. **Theorem.** *(Section 3) Let $\mathcal{R}$ be a Mackey system on $\mathfrak{K}$, and let $\mathcal{I}$ be its subcategory of injective morphisms. Then $(\mathcal{I}, \mathcal{R})$ satisfies the Green-category axioms and hence defines a Green category.*

Conversely, if a pair $(\mathcal{I}, \mathcal{R})$ of Mackey systems defines a Green category and every morphism of $\mathcal{I}$ is injective, then $\mathcal{I}$ consists precisely of the injective morphisms of $\mathcal{R}$.

Thus the induction system and the associated Green category are determined by the restriction Mackey system $\mathcal{R}$. We compare this setting with the admissible-biset framework of [BRCVE19] in Remark 4.10.

Recall that a group category is a $\mathbb{Z}$-linear subcategory of the biset category whose morphism groups have bases consisting of transitive bisets. Associated with $\mathcal{R}$ are four group categories $\mathcal{S}, \mathcal{W}, \mathcal{E}, \mathcal{N}$, with inclusions $\mathcal{S} \subseteq \mathcal{W} \subseteq \mathcal{N}$ and $\mathcal{S} \subseteq \mathcal{E} \subseteq \mathcal{N}$. Here $\mathcal{N}$ is the Green category associated with $\mathcal{R}$, $\mathcal{E}$ contains its restriction morphisms, $\mathcal{W}$ incorporates ordinary induction and restriction along surjective homomorphisms, and $\mathcal{S}$ contains the surjective-restriction morphisms. The category $\mathcal{N}$ is the inflation Mackey category of [Bar16a]. We identify the group categories with their locally unital quiver algebras, as explained in Subsection 2.C.

Let R be a commutative unital ring. As explained in Subsection 2.C, we use the same notation for the R-linear group categories and their locally unital quiver algebras, and identify functors on these categories with locally unital modules over the corresponding algebras. In this setting, we obtain the following analogue of Coşkun's Mackey structure theorem.

1.2. **Proposition.** *(Section 4) Composition induces an isomorphism of $(R\mathcal{W}, R\mathcal{E})$-bimodules*

$$R\mathcal{W} \otimes_{R\mathcal{S}} R\mathcal{E} \xrightarrow{\sim} R\mathcal{N}.$$

We also consider Green $R\mathcal{S}$-, $R\mathcal{E}$-, and $R\mathcal{N}$-functors. Their values are unital R-algebras, and their restriction maps are unital algebra homomorphisms. Green $R\mathcal{N}$-functors additionally satisfy the Frobenius relations for induction, as in [Bar16b].

Following the module-theoretic approach of Coşkun [Coş07], we realize the lower-plus construction by induction from $R\mathcal{E}$ to $R\mathcal{N}$ and the upper-plus construction by coinduction

from $R\mathcal{W}$ to $R\mathcal{N}$ after inflation from $R\mathcal{S}$ to $R\mathcal{W}$. We identify these constructions with the classical cofixed and fixed descriptions of Boltje [Bol98].

1.3. **Theorem.** *(Section 6) The lower-plus construction defines a functor*

$$(-)_+ \colon \mathsf{Fun}(R\mathcal{E}) \longrightarrow \mathsf{Fun}(R\mathcal{N})$$

left adjoint to restriction from $R\mathcal{N}$ to $R\mathcal{E}$.

The upper-plus construction defines a functor

$$(-)^+ \colon \mathsf{Fun}(R\mathcal{S}) \longrightarrow \mathsf{Fun}(R\mathcal{N})$$

such that the composite with restriction to $R\mathcal{E}$ is right adjoint to restriction from $R\mathcal{E}$ to $R\mathcal{S}$.

Both constructions preserve Green functors, and the adjunctions restrict to the corresponding categories of Green functors.

We also describe the induction and restriction maps and the multiplicative structures explicitly.

As in the classical setting, the two adjunctions yield a natural bijection

$$\mathrm{Hom}^{\mathsf{Gr}}_{R\mathcal{S}}\big({}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A), {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A)\big) \cong \mathrm{Hom}^{\mathsf{Gr}}_{R\mathcal{N}}(A_+, A^+)$$

for every Green $R\mathcal{E}$-functor A. The identity morphism on the left corresponds to the mark morphism $\mathrm{mk}^A \colon A_+ \to A^+$.

1.4. **Theorem.** *(Section 7) For every Green $R\mathcal{E}$-functor A, the mark morphism $\mathrm{mk}^A \colon A_+ \to A^+$ is a morphism of Green $R\mathcal{N}$-functors, natural in A.*

If $|G|$ is invertible in R for every $G \in \mathfrak{K}$, then mk^A is an isomorphism. Its inverse is given explicitly by Möbius inversion.

Using the Burnside ring action, we also establish an induction–restriction decomposition of $M(G)$ when $|G|$ is invertible in R. The decomposition yields a zero-detection criterion used in the proof of canonical induction; see Proposition 5.7 and Corollary 5.8.

We now turn to canonical induction. Let M be an $R\mathcal{N}$-functor and A an $R\mathcal{E}$-functor. An $R\mathcal{E}$-morphism $\nu \colon A \to {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(M)$ determines a linearization morphism $\mathrm{lin}^\nu \colon A_+ \to M$. An $R\mathcal{S}$-morphism $\pi \colon {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{N}}(M) \longrightarrow {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A)$ determines a tom Dieck morphism $\mathrm{die}^\pi \colon M \to A^+$. When the mark morphism is invertible, the associated canonical induction map is

$$\mathrm{can}^\pi = (\mathrm{mk}^A)^{-1} \circ \mathrm{die}^\pi \colon M \longrightarrow A_+.$$

Both die^π and can^π are $R\mathcal{E}$-morphisms. We give an explicit formula for can^π in Proposition 8.6. When M and A are Green functors, we also characterize its multiplicativity in Proposition 8.5.

We obtain the following analogue of Boltje's canonical induction criterion.

1.5. **Theorem.** *(Section 8) Assume that $|G|$ is invertible in R for every $G \in \mathfrak{K}$. Let M be an $R\mathcal{N}$-functor and let A be an $R\mathcal{E}$-functor. Suppose that $\nu \colon A \to {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(M)$ is an $R\mathcal{E}$-morphism and that $\pi \colon {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{N}}(M) \longrightarrow {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A)$ is an $R\mathcal{S}$-morphism. Then the following conditions are equivalent:*

(a) $\mathrm{lin}^\nu \circ \mathrm{can}^\pi = \mathrm{id}_{R\mathcal{E}\mathrm{Res}_{R\mathcal{N}}(M)}$.

(b) *For every $G \in \mathfrak{K}$ and $m \in M(G)$,*

$$\nu_G(\pi_G(m)) - m \in \sum_{H<G} {}_G\mathrm{ind}_H(M(H)).$$

In the classical setting of Mackey functors, the construction and splitting criterion recover Boltje's results [Bol98].

In the classical Mackey setting, our formula specializes over $\mathbb{Q}$ to the canonical induction formula for p-permutation modules of [BM19]. The stronger integrality result of that paper requires additional representation-theoretic arguments.

Coşkun [Coş07] also applied the module-theoretic interpretation of plus constructions to the classification of simple Mackey functors. In future work, we plan to investigate analogous questions for simple functors on Green categories and explore possible connections with the theory of simple modules over Green biset functors.

The paper is organized as follows. Section 2 contains the algebraic and biset-theoretic preliminaries. Section 3 constructs the Green category associated with a Mackey system, and Section 4 develops the four-category change-of-rings framework. Section 5 establishes the Burnside idempotent decomposition. Section 6 develops the plus constructions, while Section 7 studies the mark morphism and its inverse. Finally, Section 8 constructs the canonical induction map and establishes the splitting criterion.

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2. Preliminaries

We collect the algebraic and biset-theoretic preliminaries used throughout the paper. We first recall change-of-rings functors and their adjunctions, then the required facts about bisets and generalized biset morphisms, and finally the interpretation of group-category functors as modules over locally unital quiver algebras.

2.A. Induction, restriction and coinduction of modules.

We first recall the change-of-rings functors and adjunctions that will be used in the module-theoretic constructions below. In this subsection, all rings are unital and all modules are left modules unless otherwise stated. For a ring Γ, we write $\mathrm{Mod}(\Gamma)$ for the category of left Γ-modules and Γ-linear maps.

2.1. *Recollection.* Let $\alpha\colon \Lambda \to \Gamma$ be a unital ring homomorphism.

(a) Any Γ-module can be regarded as a Λ-module via α. This induces a functor

$$ {}_{\Lambda}\mathrm{Res}^{\alpha}_{\Gamma}\colon \mathrm{Mod}(\Gamma) \longrightarrow \mathrm{Mod}(\Lambda), $$

called restriction.

(b) Regarding Γ as a right Λ-module via α, induction is defined by

$$ {}_{\Gamma}\mathrm{Ind}^{\alpha}_{\Lambda} := \Gamma \otimes_{\Lambda} -\colon \mathrm{Mod}(\Lambda) \longrightarrow \mathrm{Mod}(\Gamma). $$

(c) Regarding Γ as a left Λ-module via α, coinduction is defined by

$$ {}_{\Gamma}\mathrm{Coind}^{\alpha}_{\Lambda} := \mathrm{Hom}_{\Lambda}(\Gamma, -)\colon \mathrm{Mod}(\Lambda) \longrightarrow \mathrm{Mod}(\Gamma). $$

We recall the standard change-of-rings adjunctions, see [Ben91, Section 2.8].

2.2. *Recollection* (Change-of-rings adjunctions). Let Λ and Γ be unital rings, and let $\alpha\colon \Lambda \to \Gamma$ be a unital ring homomorphism. Consider a Γ-module M and a Λ-module N. Then:

(a) (Adjointness of induction and restriction.) Naturally in M and N, there is an isomorphism

$$ \mathrm{Hom}_{\Gamma}\big({}_{\Gamma}\mathrm{Ind}^{\alpha}_{\Lambda}(N), M\big) \cong \mathrm{Hom}_{\Lambda}\big(N, {}_{\Lambda}\mathrm{Res}^{\alpha}_{\Gamma}(M)\big). $$

Maps $\theta\colon {}_{\Gamma}\mathrm{Ind}^{\alpha}_{\Lambda}(N) \to M$ and $\phi\colon N \to {}_{\Lambda}\mathrm{Res}^{\alpha}_{\Gamma}(M)$ correspond if and only if $\theta(\gamma \otimes n) = \gamma\phi(n)$ for all $\gamma \in \Gamma$ and $n \in N$.

(b) (Adjointness of restriction and coinduction.) Naturally in M and N, there is an isomorphism

$$\mathrm{Hom}_\Lambda\big({}_\Lambda\mathrm{Res}^\alpha_\Gamma(M), N\big) \cong \mathrm{Hom}_\Gamma\big(M, {}_\Gamma\mathrm{Coind}^\alpha_\Lambda(N)\big).$$

Maps $\phi\colon {}_\Lambda\mathrm{Res}^\alpha_\Gamma(M) \to N$ and $\psi\colon M \to {}_\Gamma\mathrm{Coind}^\alpha_\Lambda(N)$ correspond if and only if $\phi(\gamma m) = \psi(m)(\gamma)$ for all $\gamma \in \Gamma$ and $m \in M$.

2.3. *Notation.* When α is an inclusion $\Lambda \hookrightarrow \Gamma$, we omit α from the notation and write

$${}_\Gamma\mathrm{Ind}_\Lambda, \qquad {}_\Lambda\mathrm{Res}_\Gamma, \qquad {}_\Gamma\mathrm{Coind}_\Lambda\,.$$

When α is the quotient map $\Lambda \twoheadrightarrow \Gamma = \Lambda/\Delta$ for an ideal Δ of Λ, we write

$${}_\Gamma\mathrm{Def}_\Lambda \qquad \text{and} \qquad {}_\Lambda\mathrm{Inf}_\Gamma$$

for induction and restriction, respectively.

2.B. **Bisets and generalized biset morphisms.**

We next recall the biset notation and the generalized induction and restriction morphisms needed to describe the Green categories used later.

2.4. *Recollection.* Let F and G be finite groups. We write $B(F,G) = B(F \times G)$ for the Burnside group of finite (F,G)-bisets, and denote the isomorphism class of an (F,G)-biset X by $[X]$. Every transitive (F,G)-biset is isomorphic to $(F \times G)/L$ for some subgroup $L \leq F \times G$, and two transitive (F,G)-bisets are isomorphic if and only if their point stabilizers are conjugate in $F \times G$.

The biset category $\mathcal{C}$ is the $\mathbb{Z}$-linear category whose objects are finite groups and whose morphism group from G to F is $\mathcal{C}(F,G) = B(F,G)$; see [Bou10, Chapters 2, 3]. Thus an (F,G)-biset X represents the morphism $[X]\colon G \to F$ in $\mathcal{C}$. If X is an (F,G)-biset and Y is a (G,H)-biset, then $[X][Y] = [X \times_G Y]$, where $X \times_G Y$ denotes the set of G-orbits of $X \times Y$ under the action $(x,y)\cdot g = (xg, g^{-1}y)$.

For a commutative unital ring R, we write $R\mathcal{C}$ for the R-linear category obtained from $\mathcal{C}$ by extension of scalars: $R\mathcal{C}(F,G) = R \otimes_{\mathbb{Z}} B(F,G)$, with composition extended R-bilinearly.

2.5. *Definition* (Generalized induction and restriction). Given a group homomorphism $\alpha : G \to F$, we define transitive morphisms

$${}_F\mathrm{ind}^\alpha_G = \left[\frac{F \times G}{\{(\alpha(g), g) : g \in G\}}\right], \qquad {}_G\mathrm{res}^\alpha_F = \left[\frac{G \times F}{\{(g, \alpha(g)) : g \in G\}}\right]$$

called induction and restriction, respectively.

2.6. *Notation.* Following [Bar16a], we state basic results and set some notation. Let $\alpha : G \to F$ and $\beta : H \to G$ be group homomorphisms. Then

$${}_F\mathrm{ind}^\alpha_G \mathrm{ind}^\beta_H = {}_F\mathrm{ind}^{\alpha\beta}_H \qquad \text{and} \qquad {}_H\mathrm{res}^\beta_G \mathrm{res}^\alpha_F = {}_H\mathrm{res}^{\alpha\beta}_F\,.$$

If $\alpha : G \hookrightarrow F$ is injective, ${}_F\mathrm{ind}^\alpha_G$ is called ordinary induction and ${}_G\mathrm{res}^\alpha_F$ is called ordinary restriction. In this case we omit the symbol α from the notation, just writing ${}_F\mathrm{ind}_G$ and ${}_G\mathrm{res}_F$. If α is surjective, we have

$${}_F\mathrm{def}_G = {}_F\mathrm{ind}^\alpha_G \qquad \text{and} \qquad {}_G\mathrm{inf}_F = {}_G\mathrm{res}^\alpha_F\,.$$

For arbitrary α, we have the factorizations

$${}_F\mathrm{ind}^\alpha_G = {}_F\mathrm{ind}_{\alpha(G)}\mathrm{def}_G \qquad \text{and} \qquad {}_G\mathrm{res}^\alpha_F = {}_G\mathrm{inf}_{\alpha(G)}\mathrm{res}_F\,.$$

If α is an isomorphism, we set ${}_F\mathrm{iso}^\alpha_G = {}_F\mathrm{ind}^\alpha_G = {}_F\mathrm{res}^{\alpha^{-1}}_G$. If $g \in G$, then the isogation associated with the inner automorphism c_g is the identity morphism of G: ${}_G\mathrm{iso}^{c_g}_G = \mathrm{id}_G$.

2.7. *Recollection* (Generalized Mackey relation). We recall the generalized Mackey relation [Bar16b, Prop. 2.3]. For a given diagram $F \overset{\alpha}{\leftarrow} I \overset{\theta}{\to} G \overset{\beta}{\leftarrow} J \overset{\phi}{\to} H$ in the category of finite groups, we have

$$ {}_F\mathrm{ind}_I^\alpha \mathrm{res}_G^\theta \, \mathrm{ind}_J^\beta \mathrm{res}_H^\phi = \sum_{g \in \theta(I)\backslash G/\beta(J)} {}_F\mathrm{ind}_{L_g}^{\alpha\gamma_g} \mathrm{res}_H^{\phi\psi_g} \, . $$

where $I \overset{\gamma_g}{\leftarrow} L_g \overset{\psi_g}{\to} J$ is any pullback of $I \overset{\theta}{\to} G \overset{{}^g\beta}{\leftarrow} J$ where ${}^g\beta = c_g \circ \beta$.

2.8. *Recollection* (Right-free transitive morphisms). An (F, G)-biset is called *right-free* if its right G-action is free. The right-free transitive morphisms $F \leftarrow G$ are precisely the morphisms that can be expressed in the form

$$ {}_F\mathrm{ind}_V \mathrm{res}_G^\beta = {}_F\mathrm{ind}_V \mathrm{inf}_{\beta(V)}^\beta \mathrm{res}_G = \left[\frac{F \times G}{\mathfrak{G}(V, \beta)} \right] . $$

where $V \leq F$, $\beta \colon V \to G$, and $\mathfrak{G}(V, \beta) = \{(v, \beta(v)) \mid v \in V\} \leq F \times G$.

2.9. **Lemma** (Uniqueness of right-free transitive presentations). *Let $A, A' \leq G$, let $V, V' \leq F$, and let $\rho : A \twoheadrightarrow V$ and $\rho' : A' \twoheadrightarrow V'$. Suppose that*

$$ {}_G\mathrm{ind}_A \mathrm{res}_V^\rho \mathrm{res}_F = {}_G\mathrm{ind}_{A'} \mathrm{res}_{V'}^{\rho'} \mathrm{res}_F \, . $$

Then there exist $g \in G$ and $f \in F$ such that

$$ A' = {}^gA, \qquad V' = {}^fV, \qquad \rho' \circ c_g = c_f \circ \rho. $$

Proof. The two transitive bisets have stabilizers $\mathfrak{G}(A, \rho)$ and $\mathfrak{G}(A', \rho')$ in $G \times F$. By Recollection 2.4, these stabilizers are conjugate. Hence, for some $g \in G$ and $f \in F$,

$$ \mathfrak{G}(A', \rho') = {}^{(g,f)}\mathfrak{G}(A, \rho). $$

It follows that $A' = {}^gA$ and $\rho' c_g = c_f \rho$. Since ρ and ρ' are surjective, $V' = {}^fV$. □

2.C. **Group categories, functors and modules.**

Throughout this subsection, let R be a commutative unital ring. Let $\mathcal{L}$ be a small R-linear category whose objects are finite groups. By an $\mathcal{L}$-functor we mean an R-linear functor $\mathcal{L} \longrightarrow R\text{-Mod}$. The $\mathcal{L}$-functors can be regarded as modules over a locally unital algebra via the following construction. We write $\mathsf{Fun}(\mathcal{L})$ for the category of $\mathcal{L}$-functors and R-linear natural transformations. When $\mathrm{Obj}(\mathcal{L})$ is infinite, the corresponding quiver algebra need not be unital; its finite sums of identity morphisms instead form a family of local units.

2.10. *Recollection.* An R-algebra Λ, not necessarily unital, is called *locally unital* if every finite subset of Λ has a local unit, that is, an idempotent $u \in \Lambda$ such that $u\lambda = \lambda = \lambda u$ for every element λ of Λ.

A left Λ-module M is called *locally unital* if every finite subset of M is fixed by some idempotent of Λ. For an arbitrary left Λ-module M, its *locally unital part* is

$$ M_{\mathrm{lu}} = \{m \in M \mid um = m \text{ for some idempotent } u \in \Lambda\}. $$

Throughout the paper, modules over locally unital algebras are understood to be locally unital.

2.11. *Construction* (The quiver algebra). For a small R-linear category $\mathcal{L}$, define its *quiver algebra* by

$$ {}^{\oplus}\mathcal{L} = \bigoplus_{F,G \in \mathrm{Obj}(\mathcal{L})} \mathcal{L}(F, G), $$

where multiplication is given by composition when morphisms are composable and is zero otherwise.

For every finite subset $\Omega \subseteq \mathrm{Obj}(\mathcal{L})$, set $u_\Omega = \sum_{G\in\Omega} \mathrm{id}_G$. These elements form a family of local units for ${}^{\oplus}\mathcal{L}$, and every locally unital ${}^{\oplus}\mathcal{L}$-module M decomposes as

$$M = \bigoplus_{G\in\mathrm{Obj}(\mathcal{L})} \mathrm{id}_G\, M.$$

Thus, $\mathcal{L}$-functors are equivalently locally unital ${}^{\oplus}\mathcal{L}$-modules. Under this identification, an $\mathcal{L}$-functor M corresponds to $\bigoplus_{G\in\mathrm{Obj}(\mathcal{L})} M(G)$, where a morphism $x \in \mathcal{L}(F,G)$ acts on the G-summand by $M(x)\colon M(G) \to M(F)$ and annihilates the other summands. Conversely, a locally unital ${}^{\oplus}\mathcal{L}$-module M corresponds to the functor $G \mapsto \mathrm{id}_G\, M$.

We use this identification throughout when applying change-of-rings constructions. In particular, for a group category $\mathcal{X}$, we write $\mathsf{Fun}(R\mathcal{X})$ for the category of $R\mathcal{X}$-functors, identified with the category of locally unital modules over the quiver algebra $R\mathcal{X}$.

For $R\mathcal{X}$-functors A and B, we write

$$\mathrm{Hom}_{R\mathcal{X}}(A,B) := \mathrm{Hom}_{\mathsf{Fun}(R\mathcal{X})}(A,B).$$

2.12. *Recollection* (Locally unital change of rings). Let Λ and Γ be locally unital R-algebras with a common family of local units, and let $\alpha\colon \Lambda \to \Gamma$ be an R-algebra homomorphism fixing every element of this family. We consider only locally unital modules.

Restriction and induction are defined as in Recollection 2.1. For coinduction, the usual module $\mathrm{Hom}_\Lambda(\Gamma, N)$ need not be locally unital, so we take its locally unital part:

$${}_\Gamma\mathrm{Coind}^\alpha_\Lambda(N) = \mathrm{Hom}_\Lambda(\Gamma,N)_{\mathrm{lu}}.$$

With these definitions, the change-of-rings adjunctions of Recollection 2.2 remain valid. We use the notation of Notation 2.3.

2.13. *Definition* (Group categories). Following [Bar16b, Sec. 2], a *group category* $\mathcal{X}$ is a $\mathbb{Z}$-linear subcategory of the biset category $\mathcal{C}$ such that, for all $G, G' \in \mathrm{Obj}(\mathcal{X})$, the transitive morphisms in $\mathcal{X}(G',G)$ form a $\mathbb{Z}$-basis of $\mathcal{X}(G',G)$.

For a commutative unital ring R, we write $R\mathcal{X}$ for the R-linear category obtained from $\mathcal{X}$ by extension of scalars, so that $R\mathcal{X}(G',G) = R \otimes_{\mathbb{Z}} \mathcal{X}(G',G)$. By abuse of notation, we also write $R\mathcal{X}$ for its quiver algebra ${}^{\oplus}(R\mathcal{X})$.

2.14. *Notation.* For a group category $\mathcal{X}$, let $\mathrm{tm}_\mathcal{X}$ denote the set of its transitive morphisms, and set $\mathrm{tm}_\mathcal{X}(G',G) := \mathcal{X}(G',G) \cap \mathrm{tm}_\mathcal{X}$. Thus, $\mathrm{tm}_\mathcal{X}(G',G)$ is an R-basis of $R\mathcal{X}(G',G)$, and $\mathrm{tm}_\mathcal{X}$ is an R-basis of $R\mathcal{X}$. In particular,

$$R\mathcal{X} = \bigoplus_{\alpha\in\mathrm{tm}_\mathcal{X}} R\alpha.$$

3. Green categories from Mackey systems

In this section, we recall the notions of Mackey systems and Green categories that underlie our construction. Starting from a Mackey system $\mathcal{R}$, we show that its injective morphisms form a Mackey system $\mathcal{I}$ and that the pair $(\mathcal{I},\mathcal{R})$ satisfies the Green-category axioms. This framework will be applied in Construction 4.1 to construct the ambient category $\mathcal{N}$.

Let $\mathfrak{K}$ be a set of finite groups closed under taking subgroups.

3.1. *Recollection* (Mackey systems). Following [Bar16a, Sec. 3], a *Mackey system* on $\mathfrak{K}$ is a subcategory $\mathcal{F}$ of the category of finite groups such that $\mathrm{Obj}(\mathcal{F}) = \mathfrak{K}$ and, writing $\mathcal{F}(F,G)$ for the set of morphisms to a group $F \in \mathfrak{K}$ from a group $G \in \mathfrak{K}$, the following four axioms hold:

(MS 1) For every $I \le G \in \mathfrak{K}$, the inclusion $I \hookrightarrow G$ is in $\mathcal{F}(G,I)$.
(MS 2) For every $I \le G \in \mathfrak{K}$ and $g \in G$, the conjugation map $i \mapsto {}^g i$ belongs to $\mathcal{F}({}^g I, I)$.
(MS 3) For every $\theta \in \mathcal{F}(F,G)$, the corestriction $\overline{\theta}\colon G \to \theta(G)$ belongs to $\mathcal{F}(\theta(G),G)$.
(MS 4) If $\theta \in \mathcal{F}(F,G)$ is a group isomorphism, then $\theta^{-1} \in \mathcal{F}(G,F)$.

We shall use the injective morphisms in a Mackey system as a Mackey system in their own right, so we record the following elementary observation.

3.2. **Lemma.** *Let $\mathcal{R}$ be a Mackey system on $\mathfrak{K}$, and let $\mathcal{I}$ be the subcategory of $\mathcal{R}$ consisting of its injective morphisms. Then $\mathcal{I}$ is a Mackey system.*

Proof. This follows immediately from the axioms for a Mackey system. □

3.3. *Remark.* Let $\theta\colon H \to G$ belong to $\mathrm{Mor}(\mathcal{R})$, and let $\overline{\theta}\colon H \twoheadrightarrow \theta(H)$ be its corestriction. By (MS 3), $\overline{\theta} \in \mathrm{Mor}(\mathcal{R})$. If θ is injective, then $\overline{\theta}$ is a group isomorphism, and hence an isomorphism in $\mathcal{R}$ by (MS 4).

We next recall Barker's construction of Green categories from pairs of Mackey systems.

3.4. *Recollection* (Green categories). Following [Bar16b, Sec. 2], consider a pair of Mackey systems $(\mathcal{I}, \mathcal{R})$ on $\mathfrak{K}$ such that

(GC 1) The category $\mathcal{I}$ is a subcategory of $\mathcal{R}$.

(GC 2) Let $F \xleftarrow{\alpha} I \xrightarrow{\theta} G$ be a diagram with $\alpha \in \mathrm{Mor}(\mathcal{I})$ and $\theta \in \mathrm{Mor}(\mathcal{R})$. If θ is surjective and there exists a group homomorphism β such that $\alpha = \beta\theta$, then $\beta \in \mathrm{Mor}(\mathcal{I})$. If α is surjective and there exists a group homomorphism ϕ such that $\theta = \phi\alpha$, then $\phi \in \mathrm{Mor}(\mathcal{R})$.

(GC 3) For every diagram $I \xrightarrow{\theta} G \xleftarrow{\beta} J$ with $\theta \in \mathrm{Mor}(\mathcal{R})$ and $\beta \in \mathrm{Mor}(\mathcal{I})$, there is a pullback $I \xleftarrow{\gamma} L \xrightarrow{\psi} J$ with $\gamma \in \mathrm{Mor}(\mathcal{I})$ and $\psi \in \mathrm{Mor}(\mathcal{R})$.

Let $(\mathcal{I}, \mathcal{R})$ be a pair of Mackey systems satisfying the Green-category axioms. Using the induction and restriction morphisms of Definition 2.5, let $\mathcal{G}$ be the $\mathbb{Z}$-linear subcategory of $\mathcal{C}$ with $\mathrm{Obj}(\mathcal{G}) = \mathfrak{K}$ whose morphism groups are generated, as abelian groups, by finite compositions of morphisms ind^{α} and res^{θ}, where $\alpha \in \mathrm{Mor}(\mathcal{I})$ and $\theta \in \mathrm{Mor}(\mathcal{R})$.

We call $\mathcal{G}$ the *Green category* on $\mathfrak{K}$ associated with $(\mathcal{I}, \mathcal{R})$; see [Bar16b, Sec. 2]. The morphisms in $\mathcal{I}$ are called the *induction homomorphisms* for $\mathcal{G}$, and those in $\mathcal{R}$ are called the *restriction homomorphisms* for $\mathcal{G}$.

We recall the following result from [Bar16b, Prop. 2.4].

3.5. *Recollection* (Transitive morphisms in a Green category). Let $\mathcal{G}$ be the Green category associated with $(\mathcal{I}, \mathcal{R})$. Then $\mathcal{G}$ is a group category in the sense of Definition 2.13, and its transitive morphisms are precisely the morphisms $\mathrm{ind}^{\alpha}\,\mathrm{res}^{\theta}$, where $\alpha \in \mathrm{Mor}(\mathcal{I})$ and $\theta \in \mathrm{Mor}(\mathcal{R})$.

Barker [Bar16b, Ex. 2.6] considers the case in which $\mathcal{R}$ consists of all group homomorphisms and $\mathcal{I}$ of the injective ones. We now show that the same construction works for an arbitrary Mackey system $\mathcal{R}$ by taking $\mathcal{I}$ to be its subcategory of injective morphisms.

3.6. **Theorem** (Green category from a Mackey system)**.** *Let $\mathcal{R}$ be a Mackey system on $\mathfrak{K}$, and let $\mathcal{I}$ be its subcategory of injective morphisms. Then $(\mathcal{I}, \mathcal{R})$ satisfies (GC 1)–(GC 3). Consequently, the associated category of Recollection 3.4 is a Green category.*

Proof. By Lemma 3.2, $\mathcal{I}$ is a Mackey system, and (GC1) is immediate.

For (GC2), let $F \xleftarrow{\alpha} I \xrightarrow{\theta} G$ with $\alpha \in \mathrm{Mor}(\mathcal{I})$ and $\theta \in \mathrm{Mor}(\mathcal{R})$. If θ is surjective and $\alpha = \beta\theta$, then θ is injective since α is injective, hence θ is an isomorphism. Thus $\beta = \alpha\theta^{-1} \in \mathrm{Mor}(\mathcal{R})$ by (MS4), and β is injective, so $\beta \in \mathrm{Mor}(\mathcal{I})$. The other assertion is similar: if α is surjective, then α is an isomorphism and $\varphi = \theta\alpha^{-1} \in \mathrm{Mor}(\mathcal{R})$.

For (GC3), given $I \xrightarrow{\theta} G \xleftarrow{\beta} J$ with $\theta \in \mathrm{Mor}(\mathcal{R})$ and $\beta \in \mathrm{Mor}(\mathcal{I})$, let $L = \theta^{-1}(\beta(J))$. The inclusion $\gamma : L \hookrightarrow I$ lies in $\mathrm{Mor}(\mathcal{I})$ by (MS1), and there is a unique $\psi : L \to J$ with $\theta\gamma = \beta\psi$. Since $\theta\gamma \in \mathrm{Mor}(\mathcal{R})$, (MS 3) gives $\overline{\theta\gamma}\colon L \to \theta(L)$ in $\mathrm{Mor}(\mathcal{R})$. Likewise,

$\overline{\beta}\colon J \to \beta(J)$ belongs to $\mathrm{Mor}(\mathcal{R})$ and is an isomorphism, so $\overline{\beta}^{-1} \in \mathrm{Mor}(\mathcal{R})$ by (MS 4). Since $\theta(L) \leq \beta(J)$, (MS 1) gives the inclusion $\theta(L) \hookrightarrow \beta(J)$. Hence

$$\psi = \overline{\beta}^{-1} \circ \big(\theta(L) \hookrightarrow \beta(J)\big) \circ \overline{\theta\gamma} \in \mathrm{Mor}(\mathcal{R}).$$

By the definition of L and the injectivity of β, the resulting square is a pullback, proving (GC3).

Therefore $(\mathcal{I}, \mathcal{R})$ satisfies (GC1)–(GC3), and the associated category $\mathcal{G}$ is a Green category. □

3.7. **Proposition** (The injective induction system)**.** *Let $(\mathcal{I}, \mathcal{R})$ be a pair of Mackey systems satisfying (GC 1)–(GC 3). Suppose that every morphism in $\mathcal{I}$ is injective. Then $\mathcal{I}$ is precisely the subcategory of $\mathcal{R}$ consisting of its injective morphisms.*

Proof. By (GC 1), every morphism in $\mathcal{I}$ belongs to $\mathcal{R}$, and by hypothesis it is injective. Thus $\mathcal{I}$ is contained in the subcategory of injective morphisms of $\mathcal{R}$.

Conversely, let $\theta\colon H \to G$ be an injective morphism in $\mathcal{R}$, and let $\overline{\theta}\colon H \xrightarrow{\sim} \theta(H)$ be its corestriction. By (MS 3), $\overline{\theta}$ belongs to $\mathcal{R}$, and, since θ is injective, $\overline{\theta}$ is an isomorphism.

Apply (GC 2) to the diagram $H \xleftarrow{\mathrm{id}_H} H \xrightarrow{\overline{\theta}} \theta(H)$. Since $\overline{\theta}$ is surjective and $\mathrm{id}_H = \overline{\theta}^{-1}\overline{\theta}$, it follows that $\overline{\theta}^{-1} \in \mathrm{Mor}(\mathcal{I})$. Hence $\overline{\theta} \in \mathrm{Mor}(\mathcal{I})$ by (MS 4) for $\mathcal{I}$. The inclusion $\theta(H) \hookrightarrow G$ belongs to $\mathcal{I}$ by (MS 1). Thus θ, being the composite $H \xrightarrow{\overline{\theta}} \theta(H) \hookrightarrow G$, belongs to $\mathcal{I}$. □

4. A quadruple of group categories

The purpose of this section is to introduce the four group categories used in the module-theoretic description of the plus constructions. The category $\mathcal{N}$ is the Green category, while $\mathcal{E}$ contains only restriction morphisms. The category $\mathcal{W}$ contains induction together with restriction along surjective homomorphisms, and $\mathcal{S}$ is the common subcategory of $\mathcal{W}$ and $\mathcal{E}$ consisting of surjective-restriction morphisms. The main result of the section is the bimodule decomposition $\mathcal{W} \otimes_{\mathcal{S}} \mathcal{E} \cong \mathcal{N}$. This decomposition provides the change-of-rings framework for the plus constructions and their adjunctions established in Section 6.

Let $\mathcal{R}$ be a Mackey system on $\mathfrak{K}$, and let $\mathcal{I}$ be its subcategory of injective morphisms, as in Lemma 3.2.

4.A. The group categories $\mathcal{S}, \mathcal{W}, \mathcal{E}, \mathcal{N}$.

Following Coşkun's [Coş07] approach, we consider four group categories $\mathcal{S}, \mathcal{W}, \mathcal{E}, \mathcal{N}$ on $\mathfrak{K}$.

4.1. *Construction* (The four group categories). We define four $\mathbb{Z}$-linear subcategories $\mathcal{S}$, $\mathcal{W}$, $\mathcal{E}$, $\mathcal{N}$ of the biset category $\mathcal{C}$, all with object set $\mathfrak{K}$, as follows.

- $\mathcal{N}$ is generated by the morphisms ${}_G\mathrm{ind}_H^\alpha \,\mathrm{res}_F^\theta$, where $\alpha\colon H \to G$ belongs to $\mathrm{Mor}(\mathcal{I})$ and $\theta\colon H \to F$ belongs to $\mathrm{Mor}(\mathcal{R})$.
- $\mathcal{W}$ is generated by the morphisms ${}_G\mathrm{ind}_H^\alpha \,\mathrm{res}_F^\sigma$, where $\alpha\colon H \to G$ belongs to $\mathrm{Mor}(\mathcal{I})$ and $\sigma\colon H \to F$ is a surjective morphism in $\mathrm{Mor}(\mathcal{R})$.
- $\mathcal{E}$ is generated by the morphisms ${}_H\mathrm{res}_F^\theta$, where $\theta\colon H \to F$ belongs to $\mathrm{Mor}(\mathcal{R})$.
- $\mathcal{S}$ is generated by the morphisms ${}_H\mathrm{res}_F^\sigma$, where $\sigma\colon H \to F$ is a surjective morphism in $\mathrm{Mor}(\mathcal{R})$.

By Notation 2.6, the morphisms generating $\mathcal{S}$ are inflation morphisms associated with the surjective morphisms of $\mathcal{R}$.

Thus $\mathcal{E}$ contains all restriction morphisms, while $\mathcal{S}$ contains only those arising from surjective homomorphisms; similarly, $\mathcal{W}$ is obtained from $\mathcal{S}$ by adjoining induction morphisms. The resulting Green category $\mathcal{N}$ agrees with the inflation Mackey category associated with $\mathcal{R}$ in the terminology of [Bar16a, Sec. 3].

4.2. **Lemma.** *The categories $\mathcal{S}, \mathcal{W}, \mathcal{E}, \mathcal{N}$ of Construction 4.1 are group categories. Moreover, $\mathcal{S} \subseteq \mathcal{W} \subseteq \mathcal{N}$, $\mathcal{S} \subseteq \mathcal{E} \subseteq \mathcal{N}$, and $\mathcal{N}$ is the Green category associated with $(\mathcal{I}, \mathcal{R})$.*

Proof. By Construction 4.1 and Theorem 3.6, $\mathcal{N}$ is the Green category associated with $(\mathcal{I}, \mathcal{R})$. Hence, by Recollection 3.5, it is a group category, with transitive morphisms $\mathrm{ind}^{\alpha}\,\mathrm{res}^{\theta}$ for $\alpha \in \mathrm{Mor}(\mathcal{I})$ and $\theta \in \mathrm{Mor}(\mathcal{R})$.

For $\mathcal{E}$ and $\mathcal{S}$, the specified transitive morphisms are closed under composition by Notation 2.6; for $\mathcal{S}$, we also use that composites of surjective homomorphisms are surjective.

For $\mathcal{W}$, let ${}_F\mathrm{ind}_I^{\alpha}\,\mathrm{res}_G^{\sigma}$ and ${}_G\mathrm{ind}_J^{\beta}\,\mathrm{res}_H^{\tau}$ be composable transitive generators. By Recollection 2.7 and (GC 3), their composite is a sum of morphisms ${}_F\mathrm{ind}_{L_g}^{\alpha\gamma_g}\,\mathrm{res}_H^{\tau\psi_g}$, where $\gamma_g \in \mathrm{Mor}(\mathcal{I})$ and $\psi_g \in \mathrm{Mor}(\mathcal{R})$. Since $\mathcal{I}$ and $\mathcal{R}$ are categories, $\alpha\gamma_g \in \mathrm{Mor}(\mathcal{I})$ and $\tau\psi_g \in \mathrm{Mor}(\mathcal{R})$. Moreover, ψ_g is surjective, being the pullback of the surjection σ, and hence $\tau\psi_g$ is surjective. Thus every summand is again a transitive morphism of $\mathcal{W}$.

Hence, in each case, the specified transitive morphisms span the morphism groups. Since transitive bisets form a $\mathbb{Z}$-basis of the morphism groups of $\mathcal{C}$, the specified transitive morphisms form a basis. Thus $\mathcal{S}, \mathcal{W}, \mathcal{E}, \mathcal{N}$ are group categories.

Finally, ${}_H\mathrm{res}_F^{\sigma} = {}_H\mathrm{ind}_H^{\mathrm{id}_H}\,\mathrm{res}_F^{\sigma}$ gives $\mathcal{S} \subseteq \mathcal{W}$, and the remaining inclusions are immediate from Construction 4.1. □

4.3. *Remark* (Two limiting cases). The construction of Construction 4.1 includes two useful limiting cases.

If every morphism of $\mathcal{R}$ is injective, then $\mathcal{I} = \mathcal{R}$, and the ambient category $\mathcal{N}$ is the corresponding ordinary Mackey-type Green category.

At the other extreme, if $\mathcal{R}$ consists of all homomorphisms between the groups in $\mathfrak{K}$, then $\mathcal{I}$ consists of all injective homomorphisms, and $\mathcal{N}$ is Barker's category of right-free bisets; see [Bar16b, Ex. 2.6].

Thus, varying the Mackey system $\mathcal{R}$ varies the generalized restriction morphisms available in $\mathcal{E}$, and hence the ambient right-free Green category $\mathcal{N}$.

4.B. Factorizations and the $\mathcal{W} \otimes_{\mathcal{S}} \mathcal{E}$ decomposition.

The four categories are related by factorization of their transitive morphisms. We first record two such factorizations and then use the second one to identify $\mathcal{N}$ with $\mathcal{W} \otimes_{\mathcal{S}} \mathcal{E}$.

4.4. **Lemma** ($\mathcal{N}$–$\mathcal{E}$ factorization)**.** *Let $F, G \in \mathfrak{K}$ and let $\eta \in \mathrm{tm}_{\mathcal{N}}(G, F)$. Then there exist a subgroup $U \leq G$ and a transitive morphism $e \in \mathrm{tm}_{\mathcal{E}}(U, F)$ such that $\eta = {}_G\mathrm{ind}_U\, e$. Moreover, if $\eta = {}_G\mathrm{ind}_U\, e = {}_G\mathrm{ind}_V\, f$ with $e \in \mathrm{tm}_{\mathcal{E}}(U, F)$ and $f \in \mathrm{tm}_{\mathcal{E}}(V, F)$, then there exists $g \in G$ such that $V = {}^gU$ and $f = {}_V\mathrm{iso}_U^{c_g}\, e$.*

Proof. By Recollection 3.5, we may write $\eta = {}_G\mathrm{ind}_H^{\alpha}\,\mathrm{res}_F^{\theta}$ with $\alpha \in \mathrm{Mor}(\mathcal{I})$ and $\theta \in \mathrm{Mor}(\mathcal{R})$. Since α is injective, its corestriction $\overline{\alpha}\colon H \xrightarrow{\sim} \alpha(H)$ is an isomorphism in $\mathcal{R}$ by (MS 4). Hence, by Notation 2.6, $\eta = {}_G\mathrm{ind}_{\alpha(H)}\,\mathrm{res}_F^{\theta\overline{\alpha}^{-1}}$. Since $\theta\overline{\alpha}^{-1} \in \mathrm{Mor}(\mathcal{R})$, the second factor belongs to $\mathrm{tm}_{\mathcal{E}}(\alpha(H), F)$ by Construction 4.1. This proves the first assertion.

Now suppose that $\eta = {}_G\mathrm{ind}_U\, e = {}_G\mathrm{ind}_V\, f$. Write $e = {}_U\mathrm{res}_F^{\beta}$ and $f = {}_V\mathrm{res}_F^{\gamma}$, with $\beta, \gamma \in \mathrm{Mor}(\mathcal{R})$. By Recollections 2.4 and 2.8, there exist $g \in G$ and $y \in F$ such that $V = {}^gU$ and $\gamma c_g = c_y\beta$. Since inner isogations are identity morphisms by Notation 2.6, it follows that $f = {}_V\mathrm{iso}_U^{c_g}\, e$. □

The following decomposition is the analogue of Coşkun's Mackey structure theorem [Coş07, Thm. 3.2]. We first record the factorization of transitive morphisms that will be used in its proof.

4.5. *Notation.* For the remainder of this section, when forming tensor products or considering algebra structures on group categories, we regard the corresponding $\mathbb{Z}$-linear categories as their locally unital quiver algebras, as in Construction 2.11.

4.6. **Lemma** ($\mathcal{W}$ –$\mathcal{E}$ factorization)**.** *Let $\eta \in \mathrm{tm}_{\mathcal{N}}(G,F)$. Then there exist a subgroup $V \leq F$ and a transitive morphism $w \in \mathrm{tm}_{\mathcal{W}}(G,V)$ such that $\eta = w_V \mathrm{res}_F$.*

Moreover, if $\eta = w_V \mathrm{res}_F = w'_{V'} \mathrm{res}_F$, where $V, V' \leq F$, $w \in \mathrm{tm}_{\mathcal{W}}(G,V)$, and $w' \in \mathrm{tm}_{\mathcal{W}}(G,V')$, then there is an isogation $s \in \mathrm{tm}_{\mathcal{S}}(V',V)$ such that $w = w's$ and $s_V \mathrm{res}_F = {}_{V'}\mathrm{res}_F$. Consequently,

$$w \otimes_{\mathcal{S}} {}_V\mathrm{res}_F = w' \otimes_{\mathcal{S}} {}_{V'}\mathrm{res}_F\,.$$

Proof. Write $\eta = {}_G\mathrm{ind}_H^{\alpha}\,\mathrm{res}_F^{\theta}$ as in Recollection 3.5. Factoring θ through its image gives

$$\eta = \left({}_G\mathrm{ind}_H^{\alpha}\,\mathrm{res}^{\overline{\theta}}_{\theta(H)}\right)_{\theta(H)}\mathrm{res}_F,$$

where the first factor belongs to $\mathrm{tm}_{\mathcal{W}}(G,\theta(H))$.

For uniqueness, write $w = {}_G\mathrm{ind}_A\,\mathrm{res}_V^{\rho}$ and $w' = {}_G\mathrm{ind}_{A'}\,\mathrm{res}_{V'}^{\rho'}$, with $\rho\colon A \twoheadrightarrow V$ and $\rho'\colon A' \twoheadrightarrow V'$. By Lemma 2.9, there exist $g \in G$ and $f \in F$ such that $A' = {}^gA$, $V' = {}^fV$, and $\rho' c_g = c_f \rho$. Thus, for $s = {}_{V'}\mathrm{iso}_V^{c_f}$, we have $w = w's$ and $s_V \mathrm{res}_F = {}_{V'}\mathrm{res}_F$. The final equality follows from the balancing relation. □

4.7. **Proposition** (The bimodule decomposition)**.** *Composition induces an isomorphism of $(\mathcal{W},\mathcal{E})$-bimodules*

$$\Phi\colon \mathcal{W} \otimes_{\mathcal{S}} \mathcal{E} \xrightarrow{\sim} \mathcal{N}, \qquad w \otimes_{\mathcal{S}} e \longmapsto we. \tag{4.8}$$

Proof. By Lemma 4.2, composition is $\mathcal{S}$-balanced and hence induces the bimodule morphism Φ of Equation (4.8). By Lemma 4.6, every transitive morphism of $\mathcal{N}$ lies in its image, so Φ is surjective.

For $\eta \in \mathrm{tm}_{\mathcal{N}}(G,F)$, choose a factorization $\eta = w_V \mathrm{res}_F$ as in Lemma 4.6 and set

$$\Psi(\eta) = w \otimes_{\mathcal{S}} {}_V\mathrm{res}_F\,.$$

The uniqueness assertion of Lemma 4.6 shows that this is well defined. Since the transitive morphisms form a $\mathbb{Z}$-basis of $\mathcal{N}$, it extends linearly to $\Psi\colon \mathcal{N} \to \mathcal{W} \otimes_{\mathcal{S}} \mathcal{E}$. By construction, $\Phi\Psi = \mathrm{id}_{\mathcal{N}}$.

It remains to check $\Psi\Phi = \mathrm{id}$. Since the transitive morphisms span $\mathcal{W}$ and $\mathcal{E}$, it is enough to consider composable transitive morphisms $w \otimes_{\mathcal{S}} {}_K\mathrm{res}_F^{\theta}$. By Remark 3.3, ${}_K\mathrm{res}^{\overline{\theta}}_{\theta(K)} \in \mathcal{S}$. By Notation 2.6, $w_K \mathrm{res}^{\overline{\theta}}_{\theta(K)}$ is again a transitive morphism of $\mathcal{W}$. Hence

$$\Psi\Phi\left(w \otimes_{\mathcal{S}} {}_K\mathrm{res}_F^{\theta}\right) = w_K \mathrm{res}^{\overline{\theta}}_{\theta(K)} \otimes_{\mathcal{S}\theta(K)} \mathrm{res}_F = w \otimes_{\mathcal{S}} {}_K\mathrm{res}_F^{\theta}$$

by the balancing relation and Notation 2.6. Thus $\Psi\Phi = \mathrm{id}$, and the result follows. □

4.9. **Corollary.** *For every commutative unital ring R, multiplication induces an isomorphism of $(R\mathcal{W}, R\mathcal{E})$-bimodules*

$$R\mathcal{W} \otimes_{R\mathcal{S}} R\mathcal{E} \xrightarrow{\sim} R\mathcal{N}.$$

Proof. This follows from Proposition 4.7 by extension of scalars. □

4.10. *Remark* (Relation with admissible biset categories). Let $\eta = {}_G\mathrm{ind}_H^{\alpha}\,\mathrm{res}_F^{\theta} \in \mathrm{tm}_{\mathcal{N}}(G,F)$, with $\alpha \in \mathrm{Mor}(\mathcal{I})$ and $\theta \in \mathrm{Mor}(\mathcal{R})$. A point stabilizer of the corresponding transitive (G,F)-biset is $D_{\alpha,\theta} = \{(\alpha(h),\theta(h)) \mid h \in H\} \leq G \times F$. Writing p_1 and p_2 for the two projections and $k_2(D) = \{f \in F \mid (1,f) \in D\}$, we have $k_2(D_{\alpha,\theta}) = 1$, $p_1(D_{\alpha,\theta}) = \alpha(H)$, $p_2(D_{\alpha,\theta}) = \theta(H)$. Thus every transitive morphism of $\mathcal{N}$ is right-free. Moreover,

$$\begin{aligned}
\eta \in \mathrm{tm}_{\mathcal{E}}(G,F) &\iff p_1(D_{\alpha,\theta}) = G,\\
\eta \in \mathrm{tm}_{\mathcal{W}}(G,F) &\iff p_2(D_{\alpha,\theta}) = F,\\
\eta \in \mathrm{tm}_{\mathcal{S}}(G,F) &\iff p_1(D_{\alpha,\theta}) = G \text{ and } p_2(D_{\alpha,\theta}) = F.
\end{aligned}$$

This places the quadruple $(\mathcal{S},\mathcal{W},\mathcal{E},\mathcal{N})$ in the right-free part of the more general admissible-biset framework of [BRCVE19]. Their framework allows more general choices of biset operations, including deflations. However, their upper-plus construction requires

right-free bisets. Moreover, the mark morphism need not commute with deflations in general, whereas its naturality is established in the right-free setting; see [BRCVE19, Sec. 5 and Thm. 6.1]. Our framework thus concerns Green categories with injective induction homomorphisms, whose transitive morphisms are right-free.

4.C. **Module-theoretic consequences.**

Using the identification of $R\mathcal{X}$-functors with locally unital modules from Construction 2.11, we now pass from the categorical decomposition to the corresponding functor categories. The subcategory $\mathcal{S} \subseteq \mathcal{W}$ admits a complementary span generated by the transitive morphisms with nontrivial induction part. We show that this complement is an ideal, producing a projection $\mathcal{W} \to \mathcal{S}$. Together with the decomposition of the preceding subsection, this provides the restriction, induction, inflation, and coinduction maps used in the plus constructions of Section 6.

4.11. *Notation.* Using Notation 2.14, we set

$$\Delta_{\mathcal{W}} := \bigoplus_{\omega \in \mathrm{tm}_{\mathcal{W}} \setminus \mathrm{tm}_{\mathcal{S}}} \mathbb{Z}\omega.$$

Since $\mathcal{S} \subseteq \mathcal{W}$ by Lemma 4.2, we have a decomposition of $\mathbb{Z}$-modules $\mathcal{W} = \mathcal{S} \oplus \Delta_{\mathcal{W}}$.

4.12. **Lemma.** *Let $\omega = {}_G\mathrm{ind}_H^{\alpha}\,\mathrm{res}_F^{\sigma}$ be a transitive morphism in $\mathcal{W}$, where $\alpha \in \mathrm{Mor}(\mathcal{I})$ and $\sigma \in \mathrm{Mor}(\mathcal{R})$ is surjective. Then $\omega \in \mathcal{S}$ if and only if $\alpha(H) = G$. Consequently, $\Delta_{\mathcal{W}}$ is spanned by the transitive morphisms of $\mathcal{W}$ whose induction part is proper.*

Proof. By Remark 3.3 and notation 2.6, $\omega = {}_G\mathrm{ind}_{\alpha(H)}\,\mathrm{res}_F^{\sigma\overline{\alpha}^{-1}}$, where $\sigma\overline{\alpha}^{-1}$ is surjective. Thus $\alpha(H) = G$ implies $\omega \in \mathcal{S}$.

Conversely, if $\omega \in \mathcal{S}$, then its point stabilizer has first projection $\alpha(H)$, while any transitive expression of ω in $\mathcal{S}(G, F)$ has first projection G. By Recollection 2.4, these stabilizers are conjugate, so $\alpha(H) = G$.

The final assertion follows from Notation 4.11. □

4.13. **Proposition.** *The $\mathbb{Z}$-submodule $\Delta_{\mathcal{W}}$ is a two-sided ideal of $\mathcal{W}$.*

Proof. It is enough to consider composable transitive morphisms $x = {}_F\mathrm{ind}_I^{\alpha}\,\mathrm{res}_G^{\sigma}$ and $y = {}_G\mathrm{ind}_J^{\beta}\,\mathrm{res}_H^{\tau}$ in $\mathcal{W}$. As in the proof of Lemma 4.2, Recollection 2.7 expresses xy as a sum of transitive morphisms ${}_F\mathrm{ind}_{L_g}^{\alpha\gamma_g}\,\mathrm{res}_H^{\tau\psi_g}$, where $I \overset{\gamma_g}{\leftarrow} L_g \overset{\psi_g}{\to} J$ is a pullback of $I \overset{\sigma}{\to} G \overset{{}^g\beta}{\leftarrow} J$.

If $x \in \Delta_{\mathcal{W}}$, then $\alpha(I) < F$ by Lemma 4.12, and hence $(\alpha\gamma_g)(L_g) < F$ for every g. If $y \in \Delta_{\mathcal{W}}$, then ${}^g\beta(J) < G$. Since $\sigma\colon I \twoheadrightarrow G$ is surjective and ${}^g\beta(J) < G$, its inverse image $\sigma^{-1}({}^g\beta(J)) = \gamma_g(L_g)$ is a proper subgroup of I. Therefore $(\alpha\gamma_g)(L_g) < F$.

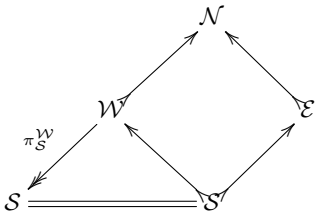
Thus every summand belongs to $\Delta_{\mathcal{W}}$ by Lemma 4.12. □

4.14. *Construction.* By Notation 4.11 and Proposition 4.13, the projection

$$\pi_{\mathcal{S}}^{\mathcal{W}}\colon \mathcal{W} \longrightarrow \mathcal{S}$$

with kernel $\Delta_{\mathcal{W}}$ is a homomorphism of locally unital $\mathbb{Z}$-algebras. Since $\pi_{\mathcal{S}}^{\mathcal{W}}$ restricts to the identity on $\mathcal{S}$, it fixes every identity morphism and hence preserves the common local units. After extension of scalars, we obtain a homomorphism $R\mathcal{W} \to R\mathcal{S}$ with kernel $R\Delta_{\mathcal{W}}$.

Thus, we have a diagram of locally unital algebra homomorphisms



where the upward arrows are inclusions.

By Construction 2.11, the four quiver algebras $R\mathcal{S}$, $R\mathcal{W}$, $R\mathcal{E}$, and $R\mathcal{N}$ have the same local units, namely the finite sums of identity morphisms.

4.15. *Notation.* The inclusions among the four group categories give restriction functors

$$ {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{W}} \colon \mathsf{Fun}(R\mathcal{W}) \longrightarrow \mathsf{Fun}(R\mathcal{S}), \qquad {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}} \colon \mathsf{Fun}(R\mathcal{E}) \longrightarrow \mathsf{Fun}(R\mathcal{S}), $$
$$ {}_{R\mathcal{W}}\mathrm{Res}_{R\mathcal{N}} \colon \mathsf{Fun}(R\mathcal{N}) \longrightarrow \mathsf{Fun}(R\mathcal{W}), \qquad {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}} \colon \mathsf{Fun}(R\mathcal{N}) \longrightarrow \mathsf{Fun}(R\mathcal{E}). $$

The projection $R\mathcal{W} \twoheadrightarrow R\mathcal{S}$ of Construction 4.14 gives the inflation functor

$$ {}_{R\mathcal{W}}\mathrm{Inf}_{R\mathcal{S}} \colon \mathsf{Fun}(R\mathcal{S}) \longrightarrow \mathsf{Fun}(R\mathcal{W}). $$

Thus, for every $R\mathcal{S}$-functor B, ${}_{R\mathcal{W}}\mathrm{Inf}_{R\mathcal{S}}(B)$ has the same underlying $R\mathcal{S}$-functor as B, with $R\Delta_{\mathcal{W}}$ acting trivially under the module identification of Construction 2.11.

4.16. **Lemma.** *There is a natural isomorphism*

$$ {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{W}}\mathrm{Inf}_{R\mathcal{S}} \cong \mathrm{id}_{\mathsf{Fun}(R\mathcal{S})}. $$

Proof. This is immediate from Notation 4.15. □

The following comparison result is the analogue of Coşkun's [Coş07, Thm. 3.4].

4.17. **Proposition** (Induction and coinduction comparisons)**.** *The following natural isomorphisms hold.*

(a) ${}_{R\mathcal{W}}\mathrm{Res}_{R\mathcal{N}}\mathrm{Ind}_{R\mathcal{E}} \cong {}_{R\mathcal{W}}\mathrm{Ind}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}$.
(b) ${}_{R\mathcal{E}}\mathrm{Coind}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{W}} \cong {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}\mathrm{Coind}_{R\mathcal{W}}$.

For an $R\mathcal{W}$-module M, under the isomorphism in (b)*, elements*

$$ \epsilon \in {}_{R\mathcal{E}}\mathrm{Coind}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{W}}(M) \quad \textit{and} \quad \nu \in {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}\mathrm{Coind}_{R\mathcal{W}}(M) $$

correspond if and only if $\nu(we) = w\epsilon(e)$ for all $w \in R\mathcal{W}$ and $e \in R\mathcal{E}$. Equivalently, ϵ is the restriction of ν to $R\mathcal{E}$.

Proof. Part (a) follows from Corollary 4.9 by tensoring over $R\mathcal{E}$.

For (b), we obtain natural $R\mathcal{E}$-isomorphisms

$$ \begin{aligned} {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}\mathrm{Coind}_{R\mathcal{W}}(M) &= \mathrm{Hom}_{R\mathcal{W}}\big({}_{R\mathcal{W}}R\mathcal{N}_{R\mathcal{E}}, M\big)_{\mathrm{lu}} && \text{by Recollection 2.12} \\ &\cong \mathrm{Hom}_{R\mathcal{W}}\big({}_{R\mathcal{W}}R\mathcal{W} \otimes_{R\mathcal{S}} R\mathcal{E}_{R\mathcal{E}}, M\big)_{\mathrm{lu}} && \text{by Corollary 4.9} \\ &\cong \mathrm{Hom}_{R\mathcal{S}}\big(R\mathcal{E}, {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{W}}(M)\big)_{\mathrm{lu}} && \text{by Recollection 2.2} \\ &= {}_{R\mathcal{E}}\mathrm{Coind}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{W}}(M) && \text{by Recollection 2.12.} \end{aligned} $$

Under these isomorphisms, ϵ corresponds to ν precisely when $\nu(we) = w\epsilon(e)$. Choosing a local unit $u \in R\mathcal{S}$ with $ue = e$, we obtain

$$ \nu(e) = \nu(ue) = u\epsilon(e) = \epsilon(e), $$

so ϵ is the restriction of ν to $R\mathcal{E}$. Conversely, if $\epsilon = \nu|_{R\mathcal{E}}$, then $\nu(we) = w\nu(e) = w\epsilon(e)$ by $R\mathcal{W}$-linearity. □

4.18. *Remark* (The triangular change-of-rings structure). The structure developed above is the triangular algebra structure considered by Coşkun; see [Coş07, pp. 235–236]. We retain its group-category realization because the cofixed and fixed descriptions of the plus constructions, the mark formula and its Möbius inverse, and the Burnside-idempotent arguments below depend essentially on the subgroup structure.

4.D. **Green functors.**

Following Barker [Bar16b, Sec. 3], we recall the notion of a Green functor for the Green category $\mathcal{N}$; see also Romero [Rom12, Sec. 2]. We also introduce Green $R\mathcal{S}$- and $R\mathcal{E}$-functors, which will be used to establish the multiplicative structures of the upper- and lower-plus constructions in Section 6.

4.19. *Definition* (Green $R\mathcal{E}$- and $R\mathcal{S}$-functors). A *Green $R\mathcal{E}$-functor* is an $R\mathcal{E}$-functor A such that:

(a) $A(G)$ is a unital R-algebra for every $G \in \mathfrak{K}$;
(b) for every $\theta\colon I \to G$ in $\mathrm{Mor}(\mathcal{R})$, the restriction map ${}_I\mathrm{res}^{\theta}_{G}\colon A(G) \longrightarrow A(I)$ is a unital R-algebra homomorphism.

A *Green $R\mathcal{S}$-functor* is defined similarly and with condition (b) required only for surjective $\theta \in \mathrm{Mor}(\mathcal{R})$.

4.20. *Definition* (Green $R\mathcal{N}$-functors). A *Green $R\mathcal{N}$-functor* is an $R\mathcal{N}$-functor A such that:

(a) $A(G)$ is a unital R-algebra for every $G \in \mathfrak{K}$;
(b) for every $\theta\colon I \to G$ in $\mathrm{Mor}(\mathcal{R})$, the restriction map ${}_I\mathrm{res}^{\theta}_{G}\colon A(G) \to A(I)$ is a unital R-algebra homomorphism;
(c) for every $\alpha\colon I \to F$ in $\mathrm{Mor}(\mathcal{I})$, $a \in A(F)$, and $b \in A(I)$, the Frobenius relations

$$ {}_F\mathrm{ind}^{\alpha}_{I}\big({}_I\mathrm{res}^{\alpha}_{F}(a)\cdot b\big) = a \cdot {}_F\mathrm{ind}^{\alpha}_{I}(b) $$

and

$$ {}_F\mathrm{ind}^{\alpha}_{I}\big(b \cdot {}_I\mathrm{res}^{\alpha}_{F}(a)\big) = {}_F\mathrm{ind}^{\alpha}_{I}(b)\cdot a $$

hold.

4.21. *Definition* (Morphisms of Green functors). Let $\mathcal{X}$ be $\mathcal{S}$, $\mathcal{E}$ or $\mathcal{N}$. A morphism $f\colon A \to B$ of Green $R\mathcal{X}$-functors is a morphism of $R\mathcal{X}$-functors such that, for every $G \in \mathfrak{K}$, the component $f_G\colon A(G) \to B(G)$ is a unital R-algebra homomorphism.

4.22. *Notation.* For $\mathcal{X} \in \{\mathcal{S},\mathcal{E},\mathcal{N}\}$, we write

$$ \mathsf{Fun}^{\mathsf{Gr}}(R\mathcal{X}) $$

for the category of Green $R\mathcal{X}$-functors and the morphisms of Definition 4.21.

For Green $R\mathcal{X}$-functors A and B, we write

$$ \mathrm{Hom}^{\mathsf{Gr}}_{R\mathcal{X}}(A,B) := \mathrm{Hom}_{\mathsf{Fun}^{\mathsf{Gr}}(R\mathcal{X})}(A,B). $$

By Lemma 4.2, Definition 4.20 is precisely Barker's definition of a Green functor for the Green category $\mathcal{N}$.

4.23. *Remark.* Restriction induces functors

$$ {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}\colon \mathsf{Fun}^{\mathsf{Gr}}(R\mathcal{E}) \longrightarrow \mathsf{Fun}^{\mathsf{Gr}}(R\mathcal{S}) \quad\text{and}\quad {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}\colon \mathsf{Fun}^{\mathsf{Gr}}(R\mathcal{N}) \longrightarrow \mathsf{Fun}^{\mathsf{Gr}}(R\mathcal{E}). $$

Consequently, ${}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{N}}$ also restricts to a functor $\mathsf{Fun}^{\mathsf{Gr}}(R\mathcal{N}) \to \mathsf{Fun}^{\mathsf{Gr}}(R\mathcal{S})$.

5. The Burnside idempotent decomposition

The purpose of this section is to separate, for an $R\mathcal{N}$-functor M, the part of $M(G)$ detected by restriction to proper subgroups from the part generated by induction from proper subgroups. We do this using the natural action of the Burnside ring on $M(G)$ and the primitive Burnside idempotent e_G^G. When $|G|$ is invertible in R, this yields an induction–restriction decomposition of $M(G)$ that will be used in the canonical-induction criterion of Section 8.

5.A. **The Burnside action.**

We begin by recalling the Burnside-ring notation that will be used throughout this section.

5.1. *Recollection.* Let $G \in \mathfrak{K}$. We write $B(G)$ for the Burnside ring of G and set $d_U^G := [G/U]$ for $U \leq G$. The elements d_U^G, with U running through representatives of the G-conjugacy classes of subgroups of G, form a $\mathbb{Z}$-basis of $B(G)$. For $U, V \leq G$, multiplication is given by

$$d_U^G \cdot d_V^G = \sum_{g \in U\backslash G/V} d_{U\cap {}^gV}^G.$$

If $H \leq G$, restriction is given by

$${}_H\mathrm{res}_G(d_U^G) = \sum_{g \in H\backslash G/U} d_{H\cap {}^gU}^H.$$

We write $RB(G) := R \otimes_{\mathbb{Z}} B(G)$.

The Burnside algebra $RB(G)$ embeds naturally in the endomorphism algebra of G.

5.2. **Proposition** (The Burnside embedding)**.** *For every $G \in \mathfrak{K}$, there is a unital injective R-algebra homomorphism $\Delta_G \colon RB(G) \longrightarrow R\mathcal{N}(G,G)$ determined by*

$$\Delta_G(d_U^G) = {}_G\mathrm{ind}_U \mathrm{res}_G$$

for $U \leq G$.

Proof. If U and V are G-conjugate, then ${}_G\mathrm{ind}_U \mathrm{res}_G = {}_G\mathrm{ind}_V \mathrm{res}_G$ by Recollection 2.4, so the assignment is well defined. Moreover, these morphisms belong to $R\mathcal{N}(G,G)$ by (MS 1) and the definition of $\mathcal{I}$.

By Recollection 2.7,

$${}_G\mathrm{ind}_U \mathrm{res}_G \mathrm{ind}_V \mathrm{res}_G = \sum_{g \in U\backslash G/V} {}_G\mathrm{ind}_{U\cap {}^gV} \mathrm{res}_G,$$

so comparison with Recollection 5.1 shows that Δ_G is multiplicative. Moreover, $\Delta_G(d_G^G) = \mathrm{id}_G$, so Δ_G is unital.

Finally, ${}_G\mathrm{ind}_U \mathrm{res}_G$ has point stabilizer $\{(u,u) \mid u \in U\} \leq G \times G$. Hence non-conjugate subgroups give distinct transitive morphisms by Recollection 2.4, and these are R-linearly independent by Notation 2.14. Thus Δ_G is injective. □

The resulting action agrees with the standard Burnside action on a Mackey functor; compare [Bol98, Proposition 1.5(i)].

5.3. *Construction* (The Burnside action). Let M be an $R\mathcal{N}$-functor and let $G \in \mathfrak{K}$. Via Proposition 5.2, we regard $M(G)$ as an $RB(G)$-module by $x \cdot m := \Delta_G(x)(m)$ for $x \in RB(G)$ and $m \in M(G)$.

In particular, for $U \leq G$, we have $d_U^G \cdot m = {}_G\mathrm{ind}_U\big({}_U\mathrm{res}_G(m)\big)$.

5.4. **Lemma** (Compatibility of the Burnside action)**.** *Let M be an $R\mathcal{N}$-functor and let $H \leq G$.*

(a) *For $x \in RB(G)$ and $m \in M(G)$, we have*

$$ {}_H\mathrm{res}_G(x \cdot m) = \big({}_H\mathrm{res}_G(x)\big) \cdot \big({}_H\mathrm{res}_G(m)\big). $$

(b) *For $x \in RB(G)$ and $n \in M(H)$, we have*

$$ x \cdot {}_G\mathrm{ind}_H(n) = {}_G\mathrm{ind}_H\left(\big({}_H\mathrm{res}_G(x)\big) \cdot n\right). $$

Proof. By R-linearity, it is enough to take $x = d_U^G$. Both assertions follow from the restriction formula in Recollection 5.1 and the generalized Mackey relation Recollection 2.7; for (b), reindex the Mackey sum by $g \mapsto g^{-1}$. □

5.B. **Primitive idempotents and the induction–restriction decomposition.**

We recall the primitive idempotents of the rational Burnside algebra; see [Bol98, Remark 2.5].

5.5. *Recollection* (Primitive Burnside idempotents). Let $G \in \mathfrak{K}$. For $H \le G$, set

$$ e_H^G := \frac{1}{|N_G(H)|} \sum_{K \le H} |K|\, \mu(K,H)\, d_K^G \in \mathbb{Q} \otimes_{\mathbb{Z}} B(G). $$

When H runs through representatives of the G-conjugacy classes of subgroups of G, the elements e_H^G form a complete set of primitive orthogonal idempotents of the rational Burnside algebra $\mathbb{Q} \otimes_{\mathbb{Z}} B(G)$.

If $|G|$ is invertible in R, then so is $|N_G(H)|$ for every $H \le G$, and the same formulas define orthogonal idempotents in $RB(G)$. In particular,

$$ e_G^G = \frac{1}{|G|} \sum_{K \le G} |K|\, \mu(K,G)\, d_K^G, $$

and therefore

$$ 1 - e_G^G = -\frac{1}{|G|} \sum_{K < G} |K|\, \mu(K,G)\, d_K^G. $$

Moreover, ${}_H\,\mathrm{res}_G(e_G^G) = 0$ for every strict subgroup $H < G$; see [Bol98, proof of Proposition 6.2].

5.6. *Notation.* Let M be an $R\mathcal{N}$-functor and let $G \in \mathfrak{K}$. Following [Bol98, Subsec. 1.6–1.7], we set

$$ \mathcal{I}(M)(G) := \sum_{H<G} {}_G\mathrm{ind}_H\big(M(H)\big) = \sum_{H<G} \mathrm{Im}\big({}_G\mathrm{ind}_H \colon M(H) \to M(G)\big) $$

and

$$ \mathcal{K}(M)(G) := \bigcap_{H<G} \mathrm{Ker}\big({}_H\mathrm{res}_G \colon M(G) \to M(H)\big). $$

The following decomposition is the analogue, for $R\mathcal{N}$-functors, of [Bol98, Proposition 6.2].

5.7. **Proposition** (The induction–restriction decomposition)**.** *Suppose that $|G|$ is invertible in R, and let M be an $R\mathcal{N}$-functor. Then*

$$ \mathcal{K}(M)(G) = \Delta_G(e_G^G) M(G), \qquad \mathcal{I}(M)(G) = \Delta_G(1 - e_G^G) M(G). $$

In particular,

$$ M(G) = \mathcal{K}(M)(G) \oplus \mathcal{I}(M)(G) $$

as $RB(G)$-modules.

Proof. By Lemma 5.4 and Recollection 5.5, $e_G^G \cdot M(G) \subseteq \mathcal{K}(M)(G)$. Conversely, if $m \in \mathcal{K}(M)(G)$, then the formula for $1 - e_G^G$ gives $(1 - e_G^G) \cdot m = 0$, and hence $\mathcal{K}(M)(G) = e_G^G \cdot M(G)$.

Similarly, the formula for $1 - e_G^G$ gives $(1 - e_G^G) \cdot M(G) \subseteq \mathcal{I}(M)(G)$. For $H < G$ and $n \in M(H)$,

$$e_G^G \cdot {}_G\mathrm{ind}_H(n) = {}_G\mathrm{ind}_H\big(({}_H\mathrm{res}_G(e_G^G)) \cdot n\big) = 0$$

by Lemma 5.4. Hence every properly induced element belongs to $(1 - e_G^G) \cdot M(G)$, so $\mathcal{I}(M)(G) = (1 - e_G^G) \cdot M(G)$.

Since $RB(G)$ is commutative and e_G^G is an idempotent, the asserted direct-sum decomposition as $RB(G)$-modules follows. □

The following zero-detection criterion is the analogue of [Bol98, Corollary 6.3].

5.8. **Corollary** (Zero detection)**.** *Suppose that $|G|$ is invertible in R. Let M be an $R\mathcal{N}$-functor and let $m \in M(G)$. Then $m = 0$ if and only if $\Delta_G(e_G^G)m = 0$ and ${}_H\mathrm{res}_G(m) = 0$ for every strict subgroup $H < G$.*

Proof. This follows immediately from Proposition 5.7. □

6. The plus constructions

This section develops the lower and upper plus constructions in the change-of-rings framework of Section 4. We identify their module-theoretic realizations with the classical cofixed and fixed descriptions, record their explicit actions and multiplicative structures, and show that the corresponding change-of-rings adjunctions restrict to Green functors.

6.A. **The lower plus construction.**

We first define the lower plus construction at the level of modules.

6.1. *Construction* (The lower-plus functor). By Lemma 4.2, we have an inclusion $R\mathcal{E} \subseteq R\mathcal{N}$. Moreover, the two locally unital algebras have the same local units by Construction 2.11.

We define the lower plus functor

$$\mathrm{Lplus} := {}_{R\mathcal{N}}\mathrm{Ind}_{R\mathcal{E}} \colon \mathsf{Fun}(R\mathcal{E}) \longrightarrow \mathsf{Fun}(R\mathcal{N}).$$

Thus, for an $R\mathcal{E}$-functor A, $\mathrm{Lplus}(A) = R\mathcal{N} \otimes_{R\mathcal{E}} A$.

We next recall the classical cofixed construction and identify it with the module-theoretic lower plus; compare [Bol98, Sec. 2.2].

6.2. *Recollection* (The cofixed lower-plus construction). Let A be an $R\mathcal{E}$-functor and let $G \in \mathfrak{K}$. The group G acts on $\bigoplus_{U \leq G} A(U)$ through the isogation maps: for $g \in G$, the summand $A(U)$ is sent to $A({}^gU)$ by ${}_gU\mathrm{iso}_U^{cg}$.

We set $A_+(G) := \Big(\bigoplus_{U \leq G} A(U)\Big)_G$. For $U \leq G$ and $x_U \in A(U)$, we write $[U, x_U]_G$ for the class represented by x_U in the U-summand. Thus

$$[{}^gU, {}_gU\,\mathrm{iso}_U^{cg}(x_U)]_G = [U, x_U]_G \qquad (g \in G).$$

Choosing one representative from each G-conjugacy class of subgroups gives an R-module isomorphism

$$A_+(G) \cong \bigoplus_{U \leq_G G} A(U)_{N_G(U)}.$$

Consequently, every element of $A_+(G)$ can be written as

$$x = \sum_{U \leq_G G} [U, x_U]_G$$

where $x_U \in A(U)$ and the class of x_U in $A(U)_{N_G(U)}$ is uniquely determined.

6.3. **Proposition** (Cofixed description of the lower plus)**.** *Let A be an $R\mathcal{E}$-functor and let $G \in \mathfrak{K}$. There is an R-linear isomorphism*

$$s_G \colon \mathrm{Lplus}(A)(G) \xrightarrow{\sim} A_+(G)$$

determined by $s_G({}_G\mathrm{ind}_U \otimes x_U) = [U, x_U]_G$ for $U \leq G$ and $x_U \in A(U)$.

Proof. Define $t_G([U, x_U]_G) := {}_G\mathrm{ind}_U \otimes x_U$. This is well defined, since for $g \in G$,

$${}_G\mathrm{ind}_{{}^gU} \otimes_{{}^gU} \mathrm{iso}_U^{c_g}(x_U) = {}_G\mathrm{ind}_{{}^gU}\mathrm{iso}_U^{c_g} \otimes x_U = {}_G\mathrm{ind}_U \otimes x_U$$

by the balancing relation and Notation 2.6.

Conversely, let $\eta \in \mathrm{tm}_{\mathcal{N}}(G, F)$. Choose a factorization $\eta = {}_G\mathrm{ind}_U e$ as in Lemma 4.4, and set $s_G(\eta \otimes x_F) := [U, e(x_F)]_G$.

The uniqueness assertion of Lemma 4.4 and the cofixed relation of Recollection 6.2 show that this is independent of the chosen factorization. Extending R-linearly, it is enough to check the balancing relation for a transitive morphism r of $\mathcal{E}$. This follows from the functoriality of A, since

$$[U, (er)(x)]_G = [U, e(r(x))]_G.$$

Finally, $s_G t_G = \mathrm{id}_{A_+(G)}$ and, if $\eta = {}_G\mathrm{ind}_U e$, then by the balancing relation,

$$t_G s_G(\eta \otimes x) = {}_G\mathrm{ind}_U \otimes e(x) = {}_G\mathrm{ind}_U e \otimes x = \eta \otimes x.$$

Thus s_G and t_G are mutually inverse. Naturality in A is immediate. □

6.4. *Construction.* Let A be an $R\mathcal{E}$-functor. For $\eta \in R\mathcal{N}(F, G)$, define

$$A_+(\eta) := s_F \circ \mathrm{Lplus}(A)(\eta) \circ s_G^{-1}.$$

Then the family $A_+(G)$ becomes an $R\mathcal{N}$-functor, and $s \colon \mathrm{Lplus}(A) \xrightarrow{\sim} A_+$ is an $R\mathcal{N}$-isomorphism.

We henceforth identify $\mathrm{Lplus}(A) \cong A_+$ via s. If $f \colon A \to B$ is a morphism of $R\mathcal{E}$-functors, we write $f_+ \colon A_+ \to B_+$ for the induced $R\mathcal{N}$-morphism. On generators,

$$f_{+,G}([U, x]_G) = [U, f_U(x)]_G.$$

6.5. **Proposition** (Explicit lower-plus action)**.** *Let A be an $R\mathcal{E}$-functor.*

(a) Let $\alpha \colon H \to G$ be a morphism in $\mathcal{I}$, let $V \leq H$, and let $y_V \in A(V)$. Writing $\overline{\alpha}_V \colon V \xrightarrow{\sim} \alpha(V)$ for the corestriction of $\alpha|_V$, we have

$${}_G\mathrm{ind}_H^{\alpha}[V, y_V]_H = [\alpha(V), {}_{\alpha(V)}\mathrm{iso}_V^{\overline{\alpha}_V}(y_V)]_G.$$

(b) Let $\theta \colon H \to G$ be a morphism in $\mathcal{R}$, let $U \leq G$, and let $x_U \in A(U)$. For $g \in \theta(H)\backslash G/U$, set $K_g := \theta^{-1}({}^gU)$. Then

$${}_H\mathrm{res}_G^{\theta}[U, x_U]_G = \sum_{g \in \theta(H)\backslash G/U} \left[K_g,\ {}_{K_g}\mathrm{res}_U^{c_{g^{-1}}\theta}(x_U)\right]_H,$$

where $c_{g^{-1}}\theta$ denotes the restriction to K_g of $c_{g^{-1}} \circ \theta$, whose image lies in U.

Proof. For (a), under the isomorphism of Proposition 6.3, the element $[V, y_V]_H$ corresponds to ${}_H\mathrm{ind}_V \otimes y_V$. By Remark 3.3, ${}_{\alpha(V)}\mathrm{iso}_V^{\overline{\alpha}_V} \in \mathcal{E}$. Hence

$${}_G\mathrm{ind}_H^{\alpha}({}_H\mathrm{ind}_V \otimes y_V) = {}_G\mathrm{ind}_{\alpha(V)}\, \mathrm{iso}_V^{\overline{\alpha}_V} \otimes y_V = {}_G\mathrm{ind}_{\alpha(V)} \otimes_{\alpha(V)} \mathrm{iso}_V^{\overline{\alpha}_V}(y_V),$$

by Notation 2.6 and the balancing relation. Applying s_G proves (a).

For (b), under the same isomorphism, $[U, x_U]_G$ corresponds to ${}_G\mathrm{ind}_U \otimes x_U$. By Recollection 2.7,

$${}_H\mathrm{res}_G^{\theta}\, \mathrm{ind}_U = \sum_{g \in \theta(H)\backslash G/U} {}_H\mathrm{ind}_{K_g}\, \mathrm{res}_U^{c_{g^{-1}}\theta},$$

by (MS 1)–(MS 3), $c_{g^{-1}}\theta\colon K_g \to U$ belongs to $\mathrm{Mor}(\mathcal{R})$, so ${}_{K_g}\mathrm{res}_U^{c_{g^{-1}}\theta} \in \mathcal{E}$. Therefore, by the balancing relation,

$$ {}_H\mathrm{res}_G^{\theta}({}_G\mathrm{ind}_U \otimes x_U) = \sum_{g\in\theta(H)\backslash G/U} {}_H\mathrm{ind}_{K_g} \otimes {}_{K_g}\mathrm{res}_U^{c_{g^{-1}}\theta}(x_U). $$

Applying s_H gives the formula in (b). □

We next record the universal property of the lower plus construction.

6.6. **Theorem** (The lower-plus construction)**.** *The lower plus functor*

$$ (-)_+\colon \mathsf{Fun}(R\mathcal{E}) \longrightarrow \mathsf{Fun}(R\mathcal{N}) $$

is left adjoint to ${}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}$*. Thus, naturally in an* $R\mathcal{E}$*-functor* A *and an* $R\mathcal{N}$*-functor* M*, there is an* R*-linear isomorphism*

$$ \mathrm{Hom}_{R\mathcal{N}}(A_+, M) \cong \mathrm{Hom}_{R\mathcal{E}}\big(A, {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(M)\big). $$

Proof. By Construction 6.1, Recollection 2.2 and Recollection 2.12, we have natural isomorphisms

$$ \mathrm{Hom}_{R\mathcal{N}}(A_+, M) \cong \mathrm{Hom}_{R\mathcal{N}}\big(R\mathcal{N}\otimes_{R\mathcal{E}} A, M\big) \cong \mathrm{Hom}_{R\mathcal{E}}\big(A, {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(M)\big). $$

Hence the lower plus functor is left adjoint to ${}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}$. □

6.7. *Remark* (Unit and counit of the lower-plus adjunction). The unit

$$ {}_+\eta_A\colon A \longrightarrow {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(A_+) $$

is, under the cofixed description of Proposition 6.3, given by ${}_+\eta_{A,G}(a) = [G, a]_G$.

The counit

$$ {}_+\varepsilon_M\colon \big({}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(M)\big)_+ \longrightarrow M $$

is given by ${}_+\varepsilon_{M,G}([U, x_U]_G) = {}_G\mathrm{ind}_U(x_U)$.

We now turn to the multiplicative structure. Assume that A is a Green $R\mathcal{E}$-functor.

6.8. **Proposition** (The lower-plus product)**.** *Let* A *be a Green* $R\mathcal{E}$*-functor. For every* $G \in \mathfrak{K}$*, there is a unital* R*-algebra structure on* $A_+(G)$ *determined by*

$$ [U, x_U]_G \cdot [V, y_V]_G = \sum_{g\in U\backslash G/V} \big[U^g \cap V,\ {}_{U^g\cap V}\mathrm{res}_U^{c_g}(x_U)\cdot {}_{U^g\cap V}\mathrm{res}_V(y_V)\big]_G, $$

where $U, V \le G$*,* $x_U \in A(U)$*, and* $y_V \in A(V)$*. The identity element is* $[G, 1_{A(G)}]_G$*.*

Proof. This is the lower-plus multiplication in the present setting. The usual double-coset argument applies; compare [BRCVE19, Theorem 7.3 and the paragraph following it]. □

6.9. **Theorem** (The Green lower plus construction)**.** *Let* A *be a Green* $R\mathcal{E}$*-functor. With the multiplication of Proposition 6.8, the lower plus functor* A_+ *is a Green* $R\mathcal{N}$*-functor.*

Moreover, the lower plus construction restricts to a functor

$$ (-)_+\colon \mathsf{Fun}^{\mathsf{Gr}}(R\mathcal{E}) \longrightarrow \mathsf{Fun}^{\mathsf{Gr}}(R\mathcal{N}). $$

Proof. By Proposition 6.8, each $A_+(G)$ is a unital R-algebra. By R-bilinearity, it is enough to verify the multiplicativity of restriction and the Frobenius relations on generators $[U, x_U]_G$. Substituting the formulas of Propositions 6.5 and 6.8 and using Recollection 2.7 gives the required identities. Moreover, Proposition 6.5(b) gives

$$ {}_I\mathrm{res}_G^{\theta}[G, 1_{A(G)}]_G = [I, 1_{A(I)}]_I, $$

so the restriction maps are unital. Hence A_+ is a Green $R\mathcal{N}$-functor. Compare the corresponding calculation in [BRCVE19, Theorem 7.3(a)].

Finally, let $f\colon A \to B$ be a morphism of Green $R\mathcal{E}$-functors. By Construction 6.4, $f_{+,G}([U, x]_G) = [U, f_U(x)]_G$. Since each f_U is a unital algebra homomorphism commuting

with restrictions, Proposition 6.8 shows that every $f_{+,G}$ is a unital algebra homomorphism. Hence f_+ is a morphism of Green $R\mathcal{N}$-functors. □

Having established the Green lower plus construction, we now record its universal property.

6.10. **Corollary** (The Green lower-plus adjunction)**.** *The adjunction of Theorem 6.6 restricts to an adjunction*

$$(-)_+ \colon \mathsf{Fun}^{\mathsf{Gr}}(R\mathcal{E}) \rightleftarrows \mathsf{Fun}^{\mathsf{Gr}}(R\mathcal{N}) \colon {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}\,.$$

Equivalently, for a Green $R\mathcal{E}$-functor A and a Green $R\mathcal{N}$-functor M, there is a natural bijection

$$\mathrm{Hom}^{\mathsf{Gr}}_{R\mathcal{N}}(A_+, M) \cong \mathrm{Hom}^{\mathsf{Gr}}_{R\mathcal{E}}\big(A, {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(M)\big).$$

Proof. By Theorem 6.9 and Remark 4.23, the two functors preserve Green functors. By Definition 4.21, it therefore remains to check that the unit and counit of Remark 6.7 are componentwise unital R-algebra homomorphisms.

The unit has this property by Proposition 6.8. For the counit, it is enough to check multiplicativity on generators. Let $x_U \in M(U)$ and $x_V \in M(V)$, and put $K_g = U^g \cap V$. By Proposition 6.8, Remark 6.7, the Frobenius relations, and the Mackey relation,

$$\begin{aligned}
{}_+\varepsilon_{M,G}\big([U, x_U]_G \cdot [V, x_V]_G\big) &= \sum_{g \in U\backslash G/V} {}_G\mathrm{ind}_{K_g}\left({}_{K_g}\mathrm{res}^{c_g}_U(x_U) \cdot {}_{K_g}\mathrm{res}_V(x_V)\right) \\
&= {}_G\mathrm{ind}_V\left({}_V\mathrm{res}_G\,\mathrm{ind}_U(x_U) \cdot x_V\right) \\
&= {}_G\mathrm{ind}_U(x_U) \cdot {}_G\mathrm{ind}_V(x_V) \\
&= {}_+\varepsilon_{M,G}\big([U, x_U]_G\big) \cdot {}_+\varepsilon_{M,G}\big([V, x_V]_G\big).
\end{aligned}$$

Thus every ${}_+\varepsilon_{M,G}$ is multiplicative. Moreover, ${}_+\varepsilon_{M,G}([G, 1_{M(G)}]_G) = 1_{M(G)}$, so it is unital. □

6.B. **The upper plus construction.**

We now develop the upper plus construction in parallel with the lower plus, beginning with its module-theoretic realization.

6.11. *Construction* (The upper-plus functor). By Lemma 4.2, we have an inclusion $R\mathcal{W} \subseteq R\mathcal{N}$. Moreover, Construction 4.14 gives a projection $R\mathcal{W} \twoheadrightarrow R\mathcal{S}$.

We define the upper plus functor

$$\mathrm{Uplus} := {}_{R\mathcal{N}}\mathrm{Coind}_{R\mathcal{W}}\mathrm{Inf}_{R\mathcal{S}} \colon \mathsf{Fun}(R\mathcal{S}) \longrightarrow \mathsf{Fun}(R\mathcal{N}).$$

Thus, for an $R\mathcal{S}$-functor B,

$$\mathrm{Uplus}(B) = {}_{R\mathcal{N}}\mathrm{Coind}_{R\mathcal{W}}\,\mathrm{Inf}_{R\mathcal{S}}(B).$$

Equivalently, by Recollection 2.12,

$$\mathrm{Uplus}(B) = \mathrm{Hom}_{R\mathcal{W}}(R\mathcal{N},\, {}_{R\mathcal{W}}\mathrm{Inf}_{R\mathcal{S}}(B))_{\mathrm{lu}}\,.$$

This is the analogue of Coşkun's realization of the classical upper plus construction as coinduction; see [Coş07, Theorem 5.2]. To relate this construction to the classical fixed description, we first identify its restriction to $R\mathcal{E}$.

6.12. **Corollary** (Restriction of the upper plus)**.** *For every $R\mathcal{S}$-functor B, there is a natural $R\mathcal{E}$-isomorphism*

$${}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}\,\mathrm{Uplus}(B) \cong {}_{R\mathcal{E}}\mathrm{Coind}_{R\mathcal{S}}(B).$$

Proof. This follows from Proposition 4.17 and Lemma 4.16. □

Although the upper plus construction is defined as an $R\mathcal{N}$-functor, Corollary 6.12 identifies its restriction to $R\mathcal{E}$ with coinduction from $R\mathcal{S}$. Consequently, the upper plus adjunction naturally takes place between $R\mathcal{E}$-functors and $R\mathcal{S}$-functors. The $R\mathcal{N}$-valued construction provides the additional induction structure needed for the mark morphism.

We next recall the classical fixed construction and identify it with the module-theoretic upper plus; compare [Bol98, Sec. 2.1].

6.13. *Recollection* (The fixed upper-plus construction). Let B be an $R\mathcal{S}$-functor and let $G \in \mathfrak{K}$. As in Recollection 6.2, the group G acts on $\bigoplus_{U\le G} B(U)$ through isogations.

We set

$$B^+(G) := \left(\bigoplus_{U\le G} B(U)\right)^G.$$

Thus an element $\xi \in B^+(G)$ is a family $\xi = (\xi^U)_{U\le G}$ with $\xi^U \in B(U)$ satisfying

$$\xi^{{}^gU} = {}_gU\mathrm{iso}_U^{c_g}(\xi^U) \qquad (g \in G). \tag{6.14}$$

Choosing one representative from each G-conjugacy class of subgroups gives an R-module isomorphism

$$B^+(G) \cong \bigoplus_{U\le_G G} B(U)^{N_G(U)}.$$

For $\xi \in B(U)^{N_G(U)}$, we write $[U,\xi]^G$ for the element of $B^+(G)$ corresponding to ξ in the U-summand of the above decomposition. Hence every $\xi \in B^+(G)$ can be written uniquely as

$$\xi = \sum_{U\le_G G} [U,\xi^U]^G, \qquad \xi^U \in B(U)^{N_G(U)}.$$

6.15. **Lemma** (Upper-plus evaluation)**.** *Let B be an $R\mathcal{S}$-functor and let $G \in \mathfrak{K}$. Evaluation at the subgroup restrictions induces a natural R-linear isomorphism*

$$\mathrm{ev}_G\colon \mathrm{Hom}_{R\mathcal{S}}\big(R\mathcal{E}\cdot\mathrm{id}_G,\, B\big) \xrightarrow{\sim} B^+(G)$$

given by $\mathrm{ev}_G(\epsilon) = \big(\epsilon({}_U\mathrm{res}_G)\big)_{U\le G}$.

Proof. Recall that by Construction 2.11, $R\mathcal{E}\cdot\mathrm{id}_G = \bigoplus_{H\in\mathfrak{K}} R\mathcal{E}(H,G)$, the left $R\mathcal{E}$-module of morphisms with source G. Let $\epsilon \in \mathrm{Hom}_{R\mathcal{S}}(R\mathcal{E}\cdot\mathrm{id}_G, B)$. For $g \in G$ and $U \le G$, we have

$$\epsilon({}_gU\mathrm{res}_G) = \epsilon\big({}_gU\mathrm{iso}_U^{c_g}\ \mathrm{res}_G\big) = {}_gU\mathrm{iso}_U^{c_g}\big(\epsilon({}_U\mathrm{res}_G)\big),$$

by Notation 2.6 and $R\mathcal{S}$-linearity. Hence $\mathrm{ev}_G(\epsilon)$ belongs to $B^+(G)$.

Conversely, let $\xi = (\xi^U)_{U\le G} \in B^+(G)$. For a transitive morphism ${}_H\mathrm{res}_G^\theta \in R\mathcal{E}\cdot\mathrm{id}_G$, we define $\epsilon_\xi({}_H\mathrm{res}_G^\theta) := {}_H\mathrm{res}_{\theta(H)}^{\overline{\theta}}\big(\xi^{\theta(H)}\big)$. This is well defined. Indeed, if ${}_H\mathrm{res}_G^\theta = {}_H\mathrm{res}_G^{\theta'}$, then, applying Lemma 2.9 to the presentations determined by the corestrictions $\overline{\theta}$ and $\overline{\theta'}$, there exist $h \in H$ and $g \in G$ such that $\theta'(H) = {}^g\theta(H)$ and $\overline{\theta'}c_h = c_g\overline{\theta}$. Using the fixed-family relation of Recollection 6.13 and Notation 2.6, we obtain

$${}_H\mathrm{res}_{\theta'(H)}^{\overline{\theta'}}\big(\xi^{\theta'(H)}\big) = {}_H\mathrm{res}_{\theta(H)}^{\overline{\theta}}\big(\xi^{\theta(H)}\big).$$

Since the transitive morphisms form an R-basis of $R\mathcal{E}\cdot\mathrm{id}_G$ by Notation 2.14, ϵ_ξ extends uniquely R-linearly to $R\mathcal{E}\cdot\mathrm{id}_G$.

It is $R\mathcal{S}$-linear. By R-linearity, it is enough to check this for a surjective $\sigma\colon K \twoheadrightarrow H$ in $\mathrm{Mor}(\mathcal{R})$ and a transitive morphism ${}_H\mathrm{res}_G^\theta$. Since σ is surjective, $(\theta\sigma)(K) = \theta(H)$, and the corestriction of $\theta\sigma$ is $\overline{\theta}\sigma$. Hence, by Notation 2.6,

$$\epsilon_\xi\big({}_K\mathrm{res}_H^\sigma\,{}_H\mathrm{res}_G^\theta\big) = \epsilon_\xi\big({}_K\mathrm{res}_G^{\theta\sigma}\big) = {}_K\mathrm{res}_{\theta(H)}^{\overline{\theta}\sigma}\big(\xi^{\theta(H)}\big) = {}_K\mathrm{res}_H^\sigma\,{}_H\mathrm{res}_{\theta(H)}^{\overline{\theta}}\big(\xi^{\theta(H)}\big) = {}_K\mathrm{res}_H^\sigma\epsilon_\xi({}_H\mathrm{res}_G^\theta).$$

Thus ϵ_ξ is $R\mathcal{S}$-linear.

Finally, for $U \leq G$, the corestriction of the inclusion $U \hookrightarrow G$ is id_U, so $\epsilon_\xi({}_U\mathrm{res}_G) = \xi^U$. Hence $\mathrm{ev}_G(\epsilon_\xi) = \xi$. Conversely, $R\mathcal{S}$-linearity gives

$$\epsilon({}_H\mathrm{res}^\theta_G) = {}_H\mathrm{res}^{\overline{\theta}}_{\theta(H)}\big(\epsilon({}_{\theta(H)}\mathrm{res}_G)\big),$$

so $\epsilon_{\mathrm{ev}_G(\epsilon)} = \epsilon$. Therefore ev_G is an isomorphism. Naturality in B is immediate. □

6.16. **Proposition** (Fixed-point description of the upper plus)**.** *Let B be an $R\mathcal{S}$-functor and let $G \in \mathfrak{K}$. There is a natural R-linear isomorphism*

$$t_G\colon \mathrm{Uplus}(B)(G) \xrightarrow{\sim} B^+(G)$$

given by

$$t_G(x) = \sum_{U \leq_G G} [U, x({}_U\mathrm{res}_G)]^G.$$

Proof. By Corollary 6.12, the component $\mathrm{Uplus}(B)(G)$ is naturally identified with

$$\mathrm{Hom}_{R\mathcal{S}}\big(R\mathcal{E}\cdot\mathrm{id}_G,\, B\big),$$

where the identification is given by restriction as in Proposition 4.17.

Now apply Lemma 6.15. It gives an R-linear isomorphism to $B^+(G)$, and its evaluation formula sends x to the invariant family whose U-coordinate is $x({}_U\mathrm{res}_G)$. Using the notation of Recollection 6.13, this is precisely

$$\sum_{U \leq_G G} [U, x({}_U\mathrm{res}_G)]^G.$$

Naturality follows from the naturality of Corollary 6.12 and Lemma 6.15. □

6.17. *Construction.* Let B be an $R\mathcal{S}$-functor. For $\eta \in R\mathcal{N}(F, G)$, define

$$B^+(\eta) := t_F \circ \mathrm{Uplus}(B)(\eta) \circ t_G^{-1}.$$

Then the family $B^+(G)$ becomes an $R\mathcal{N}$-functor, and $t\colon \mathrm{Uplus}(B) \xrightarrow{\sim} B^+$ is an $R\mathcal{N}$-isomorphism.

We henceforth write B^+ for this $R\mathcal{N}$-functor and identify $\mathrm{Uplus}(B) \cong B^+$ via t. If $f\colon B \to B'$ is a morphism of $R\mathcal{S}$-functors, we write $f^+\colon B^+ \to (B')^+$ for the induced $R\mathcal{N}$-morphism. On coordinates,

$$\big(f^+_G(\xi)\big)^U = f_U(\xi^U).$$

6.18. **Proposition** (Explicit upper-plus action)**.** *Let B be an $R\mathcal{S}$-functor.*

(a) Let $\theta\colon H \to G$ be a morphism in $\mathcal{R}$ and let $\xi \in B^+(G)$. For $V \leq H$, let $\overline{\theta}_V\colon V \twoheadrightarrow \theta(V)$ be the corestriction of $\theta|_V$. Then

$$\big({}_H\mathrm{res}^\theta_G(\xi)\big)^V = {}_V\mathrm{res}^{\overline{\theta}_V}_{\theta(V)}\big(\xi^{\theta(V)}\big).$$

(b) Let $\alpha\colon I \to F$ be a morphism in $\mathcal{I}$ and let $\eta \in B^+(I)$. For $U \leq F$ and $g \in U\backslash F/\alpha(I)$, set $K_g := \alpha^{-1}(U^g)$ and $\gamma_g := c_g\alpha|_{K_g}\colon K_g \to U$. Then

$$\big({}_F\mathrm{ind}^\alpha_I(\eta)\big)^U = \sum_{\substack{g \in U\backslash F/\alpha(I)\\ U \leq {}^g\alpha(I)}} {}_U\mathrm{iso}^{\gamma_g}_{K_g}\big(\eta^{K_g}\big).$$

For every term occurring in the sum, γ_g is an isomorphism.

Proof. Under the identification of Proposition 6.16, we regard elements of $B^+(G)$ as $R\mathcal{W}$-homomorphisms on $R\mathcal{N}\cdot\mathrm{id}_G$. The $R\mathcal{N}$-action is given by right multiplication in the argument.

For (a), if $V \leq H$, then

$$\big({}_H\mathrm{res}^\theta_G(\xi)\big)^V = \xi\big({}_V\mathrm{res}_H\,\mathrm{res}^\theta_G\big) = \xi\big({}_V\mathrm{res}^{\overline{\theta}_V}_{\theta(V)}\,\mathrm{res}_G\big) = {}_V\mathrm{res}^{\overline{\theta}_V}_{\theta(V)}\big(\xi^{\theta(V)}\big),$$

where the last equality uses $R\mathcal{W}$-linearity, since $\overline{\theta}_V$ is surjective and hence ${}_V\mathrm{res}^{\overline{\theta}_V}_{\theta(V)} \in \mathcal{S} \subseteq \mathcal{W}$. This proves (a).

Now we prove (b). For $g \in U\backslash F/\alpha(I)$, put $K_g = \alpha^{-1}(U^g)$ and $\gamma_g = c_g\alpha|_{K_g} \colon K_g \to U$. Since $K_g \hookrightarrow I$ and conjugation by g belong to $\mathrm{Mor}(\mathcal{I})$ by (MS 1) and (MS 2), Remark 3.3 and (MS 1) give $\gamma_g \in \mathrm{Mor}(\mathcal{I})$. Then

$$\begin{aligned}
\big({}_F\mathrm{ind}^\alpha_I(\eta)\big)^U &= \eta\big({}_U\mathrm{res}_F\ \mathrm{ind}^\alpha_I\big) && \text{by the coinduction action}\\
&= \sum_{g\in U\backslash F/\alpha(I)} \eta\big({}_U\mathrm{ind}^{\gamma_g}_{K_g}\mathrm{res}_I\big) && \text{by Recollection 2.7}\\
&= \sum_{g\in U\backslash F/\alpha(I)} {}_U\mathrm{ind}^{\gamma_g}_{K_g}\big(\eta({}_{K_g}\mathrm{res}_I)\big) && \text{by } R\mathcal{W}\text{-linearity of } \eta\\
&= \sum_{g\in U\backslash F/\alpha(I)} {}_U\mathrm{ind}^{\gamma_g}_{K_g}\big(\eta^{K_g}\big) && \text{by Proposition 6.16.}
\end{aligned}$$

Now $\gamma_g(K_g) = U \cap {}^g\alpha(I)$. If $U \not\leq {}^g\alpha(I)$, then $\gamma_g(K_g) < U$, so ${}_U\mathrm{ind}^{\gamma_g}_{K_g} \in \Delta_{\mathcal{W}}$ by Lemma 4.12. Since $R\Delta_{\mathcal{W}}$ acts trivially after inflation by Notation 4.15, the corresponding term vanishes.

If $U \leq {}^g\alpha(I)$, then $\gamma_g(K_g) = U$. Since γ_g is injective, it is an isomorphism, and

$${}_U\mathrm{ind}^{\gamma_g}_{K_g} = {}_U\mathrm{iso}^{\gamma_g}_{K_g}$$

by Notation 2.6. Therefore

$$\big({}_F\mathrm{ind}^\alpha_I(\eta)\big)^U = \sum_{\substack{g\in U\backslash F/\alpha(I)\\ U\leq {}^g\alpha(I)}} {}_U\mathrm{iso}^{\gamma_g}_{K_g}\big(\eta^{K_g}\big),$$

which proves (b). □

We next record the universal property of the upper plus construction.

6.19. **Theorem** (The upper-plus adjunction)**.** *The functor*

$${}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}} \circ \mathrm{Uplus}\colon \mathsf{Fun}(R\mathcal{S}) \longrightarrow \mathsf{Fun}(R\mathcal{E})$$

is right adjoint to ${}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}$*. Thus, naturally in an $R\mathcal{E}$-functor A and an $R\mathcal{S}$-functor B, there is a natural R-linear isomorphism*

$$\mathrm{Hom}_{R\mathcal{S}}\big({}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A), B\big) \cong \mathrm{Hom}_{R\mathcal{E}}\big(A, {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(B^+)\big).$$

Proof. By Corollary 6.12 and the restriction–coinduction adjunction of Recollections 2.2 and 2.12,

$$\mathrm{Hom}_{R\mathcal{S}}\big({}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A), B\big) \cong \mathrm{Hom}_{R\mathcal{E}}\big(A, {}_{R\mathcal{E}}\mathrm{Coind}_{R\mathcal{S}}(B)\big) \cong \mathrm{Hom}_{R\mathcal{E}}\big(A, {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(B^+)\big).$$

□

6.20. *Remark* (Unit and counit of the upper-plus adjunction). The unit is

$${}^+\eta_A\colon A \longrightarrow {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}\Big(\big({}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A)\big)^+\Big),$$

under the fixed description of Proposition 6.16, given by $\big({}^+\eta_{A,G}(a)\big)^U = {}_U\mathrm{res}_G(a)$.

The counit

$${}^+\varepsilon_B\colon {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{N}}(B^+) \longrightarrow B$$

is the projection onto the top coordinate: ${}^+\varepsilon_{B,G}(\xi) = \xi^G$.

We now turn to the multiplicative structure. Assume that B is a Green $R\mathcal{S}$-functor.

6.21. *Construction* (The upper-plus product). Let B be a Green $R\mathcal{S}$-functor. For $\xi, \eta \in B^+(G)$, define

$$(\xi\eta)^U := \xi^U \cdot \eta^U \qquad (U \leq G).$$

The identity element is the family $1_{B^+(G)}$ given by $(1_{B^+(G)})^U = 1_{B(U)}$ for every $U \leq G$.

The product and identity defined above satisfy the equivariance condition of Equation (6.14), since isogations are unital algebra homomorphisms. Associativity and the unit identities hold coordinatewise. Thus $B^+(G)$ is a unital R-algebra.

6.22. **Theorem** (The Green upper-plus construction)**.** *Let B be a Green $R\mathcal{S}$-functor. With the multiplication of Construction 6.21, B^+ is a Green $R\mathcal{N}$-functor.*

Moreover,

$$(-)^+ \colon \mathsf{Fun}^{\mathsf{Gr}}(R\mathcal{S}) \longrightarrow \mathsf{Fun}^{\mathsf{Gr}}(R\mathcal{N})$$

is a functor.

Proof. By Construction 6.21, each $B^+(G)$ is a unital R-algebra. The formulas of Proposition 6.18, together with the coordinatewise multiplication, give the restriction and Frobenius axioms. Hence B^+ is a Green $R\mathcal{N}$-functor; compare [BRCVE19, Theorem 7.3(b)].

Finally, let $f \colon B \to B'$ be a morphism of Green $R\mathcal{S}$-functors. By Construction 6.17,

$$\big(f^+_G(\xi)\big)^U = f_U(\xi^U).$$

Since each f_U is a unital R-algebra homomorphism, Construction 6.21 shows that every f^+_G is a unital R-algebra homomorphism. Hence f^+ is a morphism of Green $R\mathcal{N}$-functors. □

Having established the Green upper plus construction, we now record its universal property.

6.23. **Corollary** (The Green upper-plus adjunction)**.** *The adjunction of Theorem 6.19 restricts to an adjunction*

$${}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}} \colon \mathsf{Fun}^{\mathsf{Gr}}(R\mathcal{E}) \rightleftarrows \mathsf{Fun}^{\mathsf{Gr}}(R\mathcal{S}) \colon {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}} \circ (-)^+.$$

Equivalently, for a Green $R\mathcal{E}$-functor A and a Green $R\mathcal{S}$-functor B, there is a natural bijection

$$\mathrm{Hom}^{\mathsf{Gr}}_{R\mathcal{S}}\big({}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A), B\big) \cong \mathrm{Hom}^{\mathsf{Gr}}_{R\mathcal{E}}\big(A, {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(B^+)\big).$$

Proof. By Theorem 6.22 and Remark 4.23, the two functors preserve Green functors. By Definition 4.21, it remains to check that the unit and counit of Remark 6.20 are componentwise unital R-algebra homomorphisms.

The components of the unit are built from restriction maps, hence are unital algebra homomorphisms. The counit is projection onto the top coordinate, which is a unital algebra homomorphism for the coordinatewise multiplication of Construction 6.21. □

7. The mark map and its inverse

We now compare the lower and upper plus constructions. The adjunctions of Theorems 6.6 and 6.19 yield the mark map $A_+ \to A^+$. We derive its explicit formula and, when the relevant group orders are invertible in R, compute its inverse by Möbius inversion.

7.A. **Construction and basic properties.**

7.1. *Notation.* For an $R\mathcal{E}$-functor A, we write

$$A^+ := ({}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A))^+.$$

Thus A^+ denotes the upper plus construction applied to the underlying $R\mathcal{S}$-functor of A.

7.2. *Construction* (The Brauer projection). Let A be an $R\mathcal{E}$-functor and let $G \in \mathfrak{K}$. We define an R-linear map $\mathrm{br}^A\colon A_+ \longrightarrow A$ by

$$\mathrm{br}^A_G([U, x_U]_G) = \begin{cases} x_U, & U = G, \\ 0, & U < G. \end{cases}$$

Equivalently, under the decomposition of Recollection 6.2, the map br^A_G is the projection onto the summand $A(G)_{N_G(G)} = A(G)$.

By Proposition 6.5, these projections are compatible with restriction along the surjective morphisms of $\mathcal{R}$ and therefore assemble into an $R\mathcal{S}$-morphism

$$\mathrm{br}^A\colon {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{N}}(A_+) \longrightarrow {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A).$$

7.3. **Proposition** (The plus adjunction correspondence)**.** *Let A be an $R\mathcal{E}$-functor. The two plus adjunctions induce an R-linear isomorphism*

$$\Omega^A\colon \mathrm{Hom}_{R\mathcal{S}}\big({}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A), {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A)\big) \xrightarrow{\sim} \mathrm{Hom}_{R\mathcal{N}}(A_+, A^+).$$

If ϕ corresponds to $\psi = \Omega^A(\phi)$, then, for every $G \in \mathfrak{K}$, $U \le G$, and $x \in A_+(G)$,

$$\big(\psi_G(x)\big)^U = \phi_U\left(br^A_U({}_U\mathrm{res}_G(x))\right).$$

If A is a Green $R\mathcal{E}$-functor, this correspondence restricts to a bijection

$$\mathrm{Hom}^{\mathsf{Gr}}_{R\mathcal{S}}\big({}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A), {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A)\big) \cong \mathrm{Hom}^{\mathsf{Gr}}_{R\mathcal{N}}(A_+, A^+).$$

Proof. We get

$$\mathrm{Hom}_{R\mathcal{S}}\big({}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A), {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A)\big) \overset{6.19}{\cong} \mathrm{Hom}_{R\mathcal{E}}\big(A, {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(A^+)\big) \overset{6.6}{\cong} \mathrm{Hom}_{R\mathcal{N}}(A_+, A^+).$$

The composite is the asserted isomorphism Ω^A.

We now establish the coordinate formula. Let $\widetilde{\phi}\colon A \to {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(A^+)$ correspond to ϕ under the upper-plus adjunction. By Remark 6.20, $\big(\widetilde{\phi}_G(a)\big)^U = \phi_U\big({}_U\mathrm{res}_G(a)\big)$, so in particular $\big(\widetilde{\phi}_U(a)\big)^U = \phi_U(a)$.

Let $\psi = \Omega^A(\phi)$. By Remark 6.7, $\psi_U([V, a]_U) = {}_U\mathrm{ind}_V\big(\widetilde{\phi}_V(a)\big)$. Thus, by Proposition 6.18,

$$\big(\psi_U([V, a]_U)\big)^U = \begin{cases} \phi_U(a), & V = U, \\ 0, & V < U, \end{cases}$$

since $U \le {}^gV$ for some $g \in U$ only when $V = U$. Hence, by R-linearity and the definition of br^A_U,

$$\big(\psi_U(y)\big)^U = \phi_U\big(\mathrm{br}^A_U(y)\big) \qquad (y \in A_+(U)). \tag{7.4}$$

Now let $x \in A_+(G)$ and $U \le G$. Since ψ is an $R\mathcal{N}$-morphism,

$${}_U\mathrm{res}_G\big(\psi_G(x)\big) = \psi_U\big({}_U\mathrm{res}_G(x)\big).$$

Taking the U-coordinate and using Proposition 6.18 and eq. (7.4) give

$$\big(\psi_G(x)\big)^U = \phi_U\left(\mathrm{br}^A_U\big({}_U\mathrm{res}_G(x)\big)\right),$$

as claimed.

If A is Green, the last assertion follows from Corollary 6.10 and Corollary 6.23, since these are restrictions of the corresponding adjunctions for ordinary functors. □

Following the module-theoretic interpretation of [Coş07, Corollary 5.10], we define the mark map.

7.5. *Construction* (The mark map). Let A be an $R\mathcal{E}$-functor. We define the *mark map*

$$\mathrm{mk}^A\colon A_+ \longrightarrow A^+$$

to be the $R\mathcal{N}$-morphism corresponding under Proposition 7.3 to the identity morphism of ${}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A)$. Thus

$$\mathrm{mk}^A = \Omega^A\big(\mathrm{id}_{({}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A))}\big).$$

7.6. **Proposition** (The mark formula)**.** *Let A be an $R\mathcal{E}$-functor, let $G \in \mathfrak{K}$, and let $x \in A_+(G)$. For every $U \leq G$,*

$$\big(\mathrm{mk}^A_G(x)\big)^U = br^A_U({}_U\mathrm{res}_G(x)).$$

In particular, for $V \leq G$ and $x_V \in A(V)$,

$$\big(\mathrm{mk}^A_G([V,x_V]_G)\big)^U = \sum_{\substack{gV\in G/V\\ U\leq {}^gV}} {}_U\mathrm{res}_{{}^gV}\ \mathrm{iso}_V^{c_g}(x_V).$$

Proof. The first formula follows from Proposition 7.3 applied to $\phi = \mathrm{id}_{({}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A))}$, together with Construction 7.5.

For the second formula, apply Proposition 6.5 to the inclusion $U \hookrightarrow G$. We obtain

$${}_U\mathrm{res}_G[V,x_V]_G = \sum_{g\in U\backslash G/V} \Big[U\cap {}^gV,\ {}_{U\cap {}^gV}\mathrm{res}_V^{c_{g^{-1}}}(x_V)\Big]_U.$$

By Construction 7.2, only summands with $U \cap {}^gV = U$, equivalently $U \leq {}^gV$, can contribute to br^A_U. For such a g, Notation 2.6 gives

$${}_U\mathrm{res}_V^{c_{g^{-1}}} = {}_U\mathrm{res}_{{}^gV}\ \mathrm{iso}_V^{c_g}.$$

Moreover, if $U \leq {}^gV$, then $UgV = gV$. Hence the nonzero double-coset terms are in bijection with the cosets $gV \in G/V$ satisfying $U \leq {}^gV$. Substituting into the first formula proves the result. □

7.7. *Remark* (Comparison with the classical mark map). In the classical subgroup setting, the formula of Proposition 7.6 is Boltje's mark homomorphism. In particular, the maps br^A_U and mk^A_G correspond to the maps denoted π^A_U and ρ^A_G, respectively, in [Bol98, Sec. 2.3]. The module-theoretic characterization of Construction 7.5 agrees with [Coş07, Corollary 5.10].

7.8. **Theorem** (The Green mark map)**.** *Let A be a Green $R\mathcal{E}$-functor. Then*

$$\mathrm{mk}^A\colon A_+ \longrightarrow A^+$$

is a morphism of Green $R\mathcal{N}$-functors. Moreover, the maps mk^A are natural in A.

Proof. By Proposition 7.3, the correspondence Ω^A restricts to Green functors. Since the identity of ${}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A)$ is a morphism of Green $R\mathcal{S}$-functors, its image mk^A is a morphism of Green $R\mathcal{N}$-functors by Construction 7.5.

Naturality in A follows from the coordinate formula of Proposition 7.6 and the naturality of the Brauer projections. □

7.B. **The inverse mark formula.**

We now study the invertibility of the mark map, following Boltje [Bol98, Proposition 2.4]. We use the subgroup Möbius function μ on the subgroup lattice of a finite group G, as recalled in Recollection 5.5.

7.9. *Construction* (The inverse-mark operator). Let A be an $R\mathcal{E}$-functor and let $G \in \mathfrak{K}$. We define an R-linear map $\mathrm{km}^A_G\colon A^+(G) \longrightarrow A_+(G)$ by

$$\mathrm{km}^A_G(\xi) := \sum_{U\leq V\leq G} |U|\mu(U,V)\Big[U,\ {}_U\mathrm{res}_V(\xi^V)\Big]_G.$$

Here the sum runs over all pairs of subgroups $U \leq V$ of G, not over conjugacy-class representatives.

7.10. **Lemma** (Möbius inversion for the mark map)**.** *Let A be an $R\mathcal{E}$-functor and let $G \in \mathfrak{K}$. Then $\mathrm{km}_G^A \,\mathrm{mk}_G^A = |G|\,\mathrm{id}_{A_+(G)}$ and $\mathrm{mk}_G^A \,\mathrm{km}_G^A = |G|\,\mathrm{id}_{A^+(G)}$.*

Proof. For fixed G, the calculation uses only the subgroup lattice of G, the mark formula of Proposition 7.6, and the cofixed and fixed relations of Recollections 6.2 and 6.13. Thus the Möbius-inversion argument of [Bol98, Proposition 2.4] applies verbatim and gives both identities. □

7.11. **Proposition** (The inverse mark formula)**.** *Let A be an $R\mathcal{E}$-functor and let $G \in \mathfrak{K}$. Suppose that $|G|$ is invertible in R. Then*

$$\mathrm{mk}_G^A \colon A_+(G) \xrightarrow{\sim} A^+(G)$$

is an R-linear isomorphism. For every $\xi \in A^+(G)$,

$$(\mathrm{mk}_G^A)^{-1}(\xi) = \frac{1}{|G|} \sum_{U \leq V \leq G} |U| \mu(U,V) \left[U,\, {}_U\mathrm{res}_V(\xi^V)\right]_G . \tag{7.12}$$

Proof. By Lemma 7.10,

$$\mathrm{km}_G^A \,\mathrm{mk}_G^A = |G|\,\mathrm{id}_{A_+(G)} \qquad \text{and} \qquad \mathrm{mk}_G^A \,\mathrm{km}_G^A = |G|\,\mathrm{id}_{A^+(G)} .$$

Since $|G|$ is invertible in R, the map $|G|^{-1}\,\mathrm{km}_G^A$ is both a left and a right inverse of mk_G^A. The displayed formula follows from Construction 7.9. □

7.13. **Corollary** (Invertibility of the mark map)**.** *Suppose that $|G|$ is invertible in R for every $G \in \mathfrak{K}$. Then, for every $R\mathcal{E}$-functor A, the mark map*

$$\mathrm{mk}^A \colon A_+ \xrightarrow{\sim} A^+$$

is an isomorphism of $R\mathcal{N}$-functors.

If A is a Green $R\mathcal{E}$-functor, then mk^A is an isomorphism of Green $R\mathcal{N}$-functors.

Proof. By Proposition 7.11, every component mk_G^A is bijective. Since mk^A is an $R\mathcal{N}$-morphism by Construction 7.5, its componentwise inverse is automatically an $R\mathcal{N}$-morphism. Hence mk^A is an isomorphism of $R\mathcal{N}$-functors.

If A is Green, then mk^A is a morphism of Green $R\mathcal{N}$-functors by Theorem 7.8. Each component is a unital algebra isomorphism, so its inverse is also a unital algebra homomorphism. Thus the inverse is a morphism of Green $R\mathcal{N}$-functors by Definition 4.21. □

8. Canonical induction

We now study canonical induction in the setting of Green categories. We first define canonical induction as a section of the linearization map. We then use the upper-plus adjunction and the inverse mark map to construct a candidate for such a section, and determine when it is a canonical induction formula. We also study its multiplicativity.

We begin with the morphism from the lower plus construction to an $R\mathcal{N}$-functor determined by the induction–restriction adjunction.

8.1. *Construction* (The linearization map). Let A be an $R\mathcal{E}$-functor and let M be an $R\mathcal{N}$-functor. By Theorem 6.6, there is a natural R-linear isomorphism

$$\mathrm{Hom}_{R\mathcal{N}}(A_+, M) \cong \mathrm{Hom}_{R\mathcal{E}}\big(A, {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(M)\big).$$

For an $R\mathcal{E}$-morphism $\nu \colon A \to {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(M)$, we denote the corresponding $R\mathcal{N}$-morphism by

$$\mathrm{lin}^\nu \colon A_+ \longrightarrow M$$

and call it the *linearization map associated with ν*.

Under the cofixed description of Proposition 6.3, it is given by

$$\mathrm{lin}^{\nu}_{G}([U, x_U]_G) = {}_G\mathrm{ind}_U\big(\nu_U(x_U)\big)$$

for $U \leq G$ and $x_U \in A(U)$.

If A and M are Green functors and ν is a Green $R\mathcal{E}$-morphism, then lin^{ν} is a Green $R\mathcal{N}$-morphism by Corollary 6.10.

In Boltje's classical setting, when A is a restriction subfunctor of M and ν is the inclusion, lin^{ν} is the induction morphism $b^{M,A}$ of [Bol98, Sec. 3.1].

8.2. *Definition* (Canonical induction formula). Let A be an $R\mathcal{E}$-functor, let M be an $R\mathcal{N}$-functor, and let $\nu\colon A \to {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(M)$ be an $R\mathcal{E}$-morphism.

A *canonical induction formula for* lin^{ν} is an $R\mathcal{E}$-morphism

$$\alpha\colon {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(M) \longrightarrow {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(A_+)$$

such that

$$\mathrm{lin}^{\nu} \circ \alpha = \mathrm{id}_{{}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(M)}\,.$$

We next use the restriction–coinduction adjunction to construct a morphism into the upper plus construction.

8.3. *Construction* (The tom Dieck map). Let M be an $R\mathcal{N}$-functor and let A be an $R\mathcal{E}$-functor. By Theorem 6.19, there is a natural R-linear isomorphism

$$\mathrm{Hom}_{R\mathcal{S}}\big({}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{N}}(M), {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A)\big) \cong \mathrm{Hom}_{R\mathcal{E}}\big({}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(M), {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(A^+)\big).$$

Under this isomorphism, given an $R\mathcal{S}$-morphism $\pi\colon {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{N}}(M) \longrightarrow {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A)$, there exists a corresponding $R\mathcal{E}$-morphism $\mathrm{die}^{\pi}\colon {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(M) \longrightarrow {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(A^+)$ which we call the *tom Dieck map associated with* π.

Under the fixed description of Proposition 6.16, its coordinates are

$$\big(\mathrm{die}^{\pi}_{G}(m)\big)^{U} = \pi_U\big({}_U\mathrm{res}_G(m)\big)$$

for $U \leq G$. Equivalently,

$$\mathrm{die}^{\pi}_{G}(m) = \sum_{U \leq_G G} \big[U,\, \pi_U\big({}_U\mathrm{res}_G(m)\big)\big]^G\,.$$

Indeed, the top coordinate is given by Remark 6.20, and the remaining coordinates follow from restriction compatibility and Proposition 6.18.

We now use the inverse mark map to construct a candidate for a canonical induction formula. By Corollary 7.13, the mark map is invertible whenever $|G|$ is invertible in R for every $G \in \mathfrak{K}$.

8.4. *Construction* (The canonical induction map). Let M be an $R\mathcal{N}$-functor and let A be an $R\mathcal{E}$-functor. Suppose that

$$\mathrm{mk}^{A}\colon A_+ \longrightarrow A^+$$

is an isomorphism of $R\mathcal{N}$-functors, and let $\pi\colon {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{N}}(M) \longrightarrow {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A)$ be an $R\mathcal{S}$-morphism. We define the *canonical induction map associated with* π to be the composite

$$\mathrm{can}^{\pi} := (\mathrm{mk}^{A})^{-1} \circ \mathrm{die}^{\pi}\,.$$

Thus

$$\mathrm{can}^{\pi}\colon {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(M) \longrightarrow {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(A_+)$$

is an $R\mathcal{E}$-morphism.

Equivalently, after suppressing the restriction symbols, we have the commutative triangle

$$\begin{array}{ccc} A_+ & \xrightarrow{\ \mathrm{mk}^{A}\ } & A^+ \\ & {\scriptstyle \mathrm{can}^{\pi}} \nwarrow \quad \nearrow {\scriptstyle \mathrm{die}^{\pi}} & \\ & M & \end{array}$$

of $R\mathcal{E}$-functors.

We now characterize when the canonical induction map is a morphism of Green functors.

8.5. **Proposition** (Multiplicativity of canonical induction)**.** *Let M be a Green $R\mathcal{N}$-functor and let A be a Green $R\mathcal{E}$-functor. Suppose that $\mathrm{mk}^A\colon A_+ \xrightarrow{\sim} A^+$ is an isomorphism of Green $R\mathcal{N}$-functors, and let $\pi\colon {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{N}}(M) \to {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A)$ be an $R\mathcal{S}$-morphism. Then the following conditions are equivalent:*

(a) π is a Green $R\mathcal{S}$-morphism;
(b) die^π is a Green $R\mathcal{E}$-morphism;
(c) can^π is a Green $R\mathcal{E}$-morphism.

Proof. The equivalence of the first two conditions follows from Corollary 6.23, since the ordinary upper-plus adjunction of Theorem 6.19 restricts to Green functors.

Since mk^A is an isomorphism of Green $R\mathcal{N}$-functors, both mk^A and $(\mathrm{mk}^A)^{-1}$ restrict to morphisms of Green $R\mathcal{E}$-functors by Remark 4.23. Hence die^π is a Green $R\mathcal{E}$-morphism if and only if $\mathrm{can}^\pi = (\mathrm{mk}^A)^{-1} \circ \mathrm{die}^\pi$ is one. □

For the explicit formula and the splitting criterion below, we return to the hypothesis that $|G|$ is invertible in R for every $G \in \mathfrak{K}$. The inverse mark formula gives an explicit description of the canonical induction map.

8.6. **Proposition** (Explicit canonical induction formula)**.** *Assume that $|G|$ is invertible in R for every $G \in \mathfrak{K}$. Let M, A, and π be as in Construction 8.4. For $G \in \mathfrak{K}$ and $m \in M(G)$, one has*

$$\mathrm{can}^\pi_G(m) = \frac{1}{|G|} \sum_{U \le V \le G} |U|\mu(U,V)\,[U,\, {}_U\mathrm{res}_V\,(\pi_V({}_V\mathrm{res}_G(m)))]_G\,.$$

Here the sum runs over all pairs of subgroups $U \le V$ of G.

Proof. Set $\xi = \mathrm{die}^\pi_G(m)$. By Construction 8.3,

$$\xi^V = \pi_V\big({}_V\mathrm{res}_G(m)\big) \qquad (V \le G).$$

Since $\mathrm{can}^\pi_G(m) = (\mathrm{mk}^A_G)^{-1}(\xi)$, the result follows immediately from Proposition 7.11. □

In Boltje's classical setting, this specializes to [Bol98, Eq. (6.1a)]. In classical canonical induction formulas, the projections onto linear characters or exprojective p-permutation modules need not preserve multiplication; see [Bol98, BM19]. Consequently, the corresponding canonical induction maps need not be morphisms of Green functors.

We now determine when the map can^π is a canonical induction formula for lin^ν. The following criterion generalizes [Bol98, Proposition 6.4].

8.7. **Theorem** (Canonical induction criterion)**.** *Assume that $|G|$ is invertible in R for every $G \in \mathfrak{K}$. Let M be an $R\mathcal{N}$-functor and let A be an $R\mathcal{E}$-functor. Suppose that we are given an $R\mathcal{E}$-morphism $\nu\colon A \to {}_{R\mathcal{E}}\mathrm{Res}_{R\mathcal{N}}(M)$ and an $R\mathcal{S}$-morphism $\pi\colon {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{N}}(M) \to {}_{R\mathcal{S}}\mathrm{Res}_{R\mathcal{E}}(A)$. Then the following conditions are equivalent:*

(a) can^π is a canonical induction formula for lin^ν.
(b) For every $G \in \mathfrak{K}$ and $m \in M(G)$,

$$\nu_G(\pi_G(m)) - m \in \sum_{H<G} {}_G\mathrm{ind}_H(M(H)).$$

Proof. The argument is the same as that of [Bol98, Prop. 6.4]; we indicate the points needed in the present setting.

By Proposition 8.6 and Construction 8.1 and Lemma 5.4,

$$e^G_G \cdot \big(\mathrm{lin}^\nu_G\,\mathrm{can}^\pi_G(m)\big) = e^G_G \cdot \nu_G(\pi_G(m))$$

for every $G \in \mathfrak{K}$ and $m \in M(G)$, since e_G^G annihilates every term induced from a proper subgroup.

If $\mathrm{lin}^\nu \circ \mathrm{can}^\pi = \mathrm{id}_M$, the displayed identity and Proposition 5.7 give

$$\nu_G(\pi_G(m)) - m \in \mathcal{I}(M)(G).$$

Conversely, assume (b). Since lin^ν is an $R\mathcal{N}$-morphism and can^π is an $R\mathcal{E}$-morphism, the composite $\mathrm{lin}^\nu \circ \mathrm{can}^\pi$ is an $R\mathcal{E}$-endomorphism and hence commutes with restriction to subgroups. The induction argument of [Bol98, Prop. 6.4], using Corollary 5.8, therefore gives $\mathrm{lin}^\nu_G \mathrm{can}^\pi_G = \mathrm{id}_{M(G)}$ for every $G \in \mathfrak{K}$. □

8.8. *Remark* (The p-permutation formula). The canonical induction formula of [BM19] is recovered from Proposition 8.6 and Theorem 8.7 in the classical Mackey setting, where $\mathcal{R}$ consists of injective homomorphisms. In this case, $\mathcal{S}$ contains only isogations.

Take

$$M(G) = \mathbb{Q}T(G), \qquad A(G) = \mathbb{Q}T^{\mathrm{ex}}(G),$$

where $T(G)$ is the p-permutation ring and $T^{\mathrm{ex}}(G)$ is its subring spanned by the classes of exprojective modules. The inclusion $\nu\colon A \longrightarrow M$ is a morphism of Green restriction functors, while π_G is the projection onto $\mathbb{Q}T^{\mathrm{ex}}(G)$ along the indecomposable non-exprojective classes. Since exprojectivity is preserved under isogation, the projections π_G commute with isogation maps. Thus $\pi = (\pi_G)_{G\in\mathfrak{K}}$ is an $R\mathcal{S}$-morphism.

Then Proposition 8.6 gives the map of [BM19, Theorem 3.1], while Theorem 8.7 reduces its splitting property to the proper-induction condition verified in the proof of that theorem. The stronger $\mathbb{Z}[1/p]$-integrality statement requires the additional representation-theoretic argument of [BM19, Theorem 3.4].

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